

Quotient Atlases in Representation Learning: Region-Varying Quotient Structure and Its Dimension Floor

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Abstract

Many representation-learning problems exhibit equivalence structure that varies across the input domain rather than holding globally. The phase space of an ideal Hamiltonian double pendulum stratifies into four topologically distinct energy regimes separated by Morse-critical values. The HVAC control system of a passenger-ferry ship-hotel runs through four thermal regimes determined by threshold switches in cabin and car-bay temperatures, with no smooth continuum between them. In each case the natural object is not a single quotient $\mathcal{X} \rightarrow Q$ but a family of local quotient structures indexed by a base, with different local quotients connected by compatibility data of variable strength. We call such structures *quotient atlases* and develop their theory.

Three tiers are distinguished by what is supplied as input. At the *given-partition* tier, the atlas is specified and the task reduces to per-region quotient-faithful encoding plus region-recognition. For partition atlases at the homogeneous topological-Euclidean contract, we prove a partition base-split atlas dimension-floor theorem: any base-split atlas-faithful encoder into an $\mathbb{R}^{D_{\text{ambient}}}$ latent ambient satisfies $D_{\text{ambient}} \geq d_B + \sup_{b \in B} \text{emb}(Q_b)$, the two terms reading as the base-embedding cost and the hardest local-quotient embedding cost. The base-split hypothesis is a representation-contract choice: it names a coordinate splitting in the latent ambient that recovers the base coordinate independently of fibre content. Encodings that entangle base and fibre information in the same latent coordinates fall outside the contract and are carried as a named open problem rather than a defect of the bound. Open-overlap and stratified-meet versions of the bound, as well as singular transitions and stronger contracts, are likewise carried as named open problems rather than absorbed into a bundled additive cost.

At the *learned-partition* tier, the atlas is drawn from an admissible family specified by per-region structural specifications and a regime-indicator class. We describe joint selection, evidence criteria, and the evidence-equivalence-class phenomenon that makes atlas hypotheses generically non-unique under selection. At the *emergent* tier, no prior family is supplied and the partition and per-region structure must be co-discovered from data. We do not claim this is solved. We identify a three-component structural gap (faithful-summary object, base poset or site, training-dynamics compatibility), present a portfolio of diagnostic tool classes at different readiness levels, and state nineteen open problems.

The theorem assumes memoryless compatibility; hysteretic and direction-dependent overlap structures are excluded from the present statement. Local quotients and bases

are assumed integer-dimensional.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series *Quotient Faithfulness in Representation Learning* develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.

Paper I [Farkas, 2026a]: Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.

Paper II [Farkas, 2026b]: Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.

Paper III [Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.

Paper IV [Farkas, 2026d] (this paper): Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.

Paper V [Farkas, 2026e]: Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.

Paper VI [Farkas, 2026f]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.

Paper VII [Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].

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1 Introduction

Many representation-learning problems do not admit a single global equivalence relation on their input space. Two systems make the point.

Double pendulum. The configuration space of a double pendulum is the 2-torus T^2 and its full phase space is $T^*T^2 \cong T^2 \times \mathbb{R}^2$. Gravity breaks the $U(1) \times U(1)$ symmetry of the configuration torus; the system is non-integrable; its phase-space topology stratifies by energy. For an ideal Hamiltonian double pendulum (equal links, equal masses, conservative dynamics, compact energy truncation) the regular energy surfaces are $\emptyset \rightarrow S^3 \rightarrow S^2 \times S^1 \rightarrow \#_2(S^2 \times S^1) \rightarrow T^3$ as energy crosses four Morse-critical values, by stratified Morse theory on the sublevel sets [Milnor, 1963a, Smale, 1970a]. Unequal masses or links shift the critical energies without changing the qualitative ordering; non-conservative perturbations (damping, forcing, joint limits, control laws) invalidate the Hamiltonian stratification entirely. The dynamical content is not a single quotient $\mathcal{X} \rightarrow Q$ but a family of local quotient structures indexed by energy regime, each with its own fibre topology and dynamical character.

Color Fantasy ship-hotel HVAC. A passenger-ferry HVAC system controls cabin air temperature against ambient heat loss, with four thermal regimes — cooling, neutral, moderate heating, strong heating — separated by hard threshold switches on the internal-temperature setpoint. Outdoor temperature and windchill jointly drive the heat loss; the setpoint responds; the active regime is determined by the joint state of cabin and car-bay temperatures. Each regime has its own thermal-mass delay structure, its own operative subset of control variables, and its own regression relationship between inputs and energy consumption. A day–night cyclic variable $\varphi_{\text{day}} \in S^1$ is shared across regimes. Regime assignment is determined by state, switches are discrete, and there is no smooth continuum between regimes. The system is documented in Farkas [2026g] as part of an industrial deployment study.

In each case, treating the problem as a single global quotient either fails outright or applies only within a region. Task performance, trained over the full domain, can be achieved by encoders that pass within-region evaluation while assigning different unfaithful encodings in different regions — a failure mode undetectable from within-region evaluation alone.

What this paper does. This paper develops the theory of *quotient atlases*: input spaces equipped with a region decomposition over a base, per-region local quotient structures, and compatibility data on overlaps. The theory generalises the single-quotient frameworks of earlier work in this series; results are cited where they are inherited from or extended, but the paper is intended to be readable on its own terms. The scope is named plainly: only integer-dimension local quotients and bases, memoryless compatibility, and a floor theorem at the homogeneous topological-Euclidean contract under a base-split hypothesis on the encoder. Each restriction is a named future-work direction (§7) rather than a claim about how the theory ought to develop.

What this paper is and is not. The paper does not solve general quotient-atlas embedding. It isolates the clean base-split case, proves the corresponding additive floor, and uses the failure modes of that theorem to organise the harder atlas theory. Theorem 4.8 is a floor for encoders that allocate independent latent coordinates to the base and to fibre content; the strength of the bound comes from the strength of the base-split

hypothesis. The substantive contribution is the framework: the distinction between base, regime labels, and fibre objects; the compatibility-strength axis; the three-tier separation between given, learned, and emergent atlas hypotheses; and the open problems that organise what remains. Whether the base-split bound is also a general atlas-faithful bound — whether non-base-split encoders can ever achieve strictly lower latent dimensions for atlas-structured problems — is open (§7.3.1).

Structure of the paper. §2 works the two motivating systems in enough detail to establish that atlas structure is a genuine phenomenon with at least two distinct flavours. §3 gives definitions and introduces the compatibility-strength axis along which atlas structures vary, with local models for transitions across regime boundaries. §4 develops the given-partition tier and proves the partition base-split atlas dimension-floor theorem at the topological contract. §5 develops the learned-partition tier; it inherits generator machinery from earlier work in the series where applicable and supplements it with non-Lie-type per-regime specifications where that machinery does not apply. §6 discusses emergent discovery: structural gap, diagnostic tool classes, readiness. §7 lists open problems.

The paper does not claim that emergent atlas discovery is solved. It does claim that the distinction between the three tiers is useful, that the given and learned tiers admit a clean quotient-theoretic formulation, and that the emergent tier is where the hardest and most valuable open questions live.

1.1 Notation and vocabulary

This subsection collects the symbols used in the rest of the paper, organised by topic. The reader is encouraged to scan the topic headings rather than read end-to-end.

Standard mathematical objects.

Symbol	Meaning
$\mathbb{N}$	Natural numbers (including 0).
$\mathbb{Z}$	Integers.
$\mathbb{Q}$	Rational numbers.
$\mathbb{R}$	Real numbers.
$\mathbb{R}^d$	d -dimensional Euclidean space.
$\mathbb{Q}_p$	The field of p -adic numbers (§3.5, §7.1.5).
S^d	The d -sphere.
T^d	The d -torus.
$\mathbb{CP}^d$	Complex projective space of complex dimension d (§7.1.3).
$\hookrightarrow$	Topological (or stronger) embedding.
$\rightarrow$	Continuous map.
$\twoheadrightarrow$	Quotient map.
$\cong$	Homeomorphism (or isomorphism in the appropriate category).
$\simeq$	Homotopy equivalence.
$\#$	Connected sum (§2.2).

Typographic conventions. Where a symbol is introduced, its role can usually be inferred from its style:

- *Lowercase italic* (b , α , d_B) — points, labels, integers, scalar quantities.

- *Uppercase italic* (B, R_α, E, Q_b) — sets, spaces, maps between spaces, structured objects.
- *Calligraphic uppercase* ($\mathcal{X}, \mathcal{Q}, \mathcal{T}_\alpha, \mathcal{H}, \mathcal{N}$) — distinguished objects of higher complexity: the input space, the atlas itself, classes of spaces, function spaces, simplicial complexes.
- *Blackboard bold* ($\mathbb{R}, \mathbb{N}, \mathbb{Q}_p$) — number systems and their canonical Euclidean spaces.
- *Greek lowercase* (α, ι, ρ, π) — labels, indices, named functions.
- *Greek uppercase* (Λ, Σ_E) — multifunctions and named subspaces with specific roles.
- *Subscripts on functions* ($\pi_B, \pi_\alpha, \tilde{E}_\alpha$) — indicate source or domain of restriction.
- *Tildes* ($\tilde{E}_\alpha$) — derived or factored versions of a base symbol.

The data–network chain. The objects relating data, encoder, and architecture sit in a chain:

space $\rightarrow$ training data $\rightarrow$ latent shape $\rightarrow$ latent ambient $\rightarrow$ function space $\rightarrow$ parameter realisation,
following Paper I.

Symbol	Concept	Type and notes
$\mathcal{X}$	The space	Compact Hausdorff topological space — typically a manifold, an atlas of manifolds, or a stratified or singular space.
E	The encoder	Continuous map $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$.
$E(\mathcal{X})$	The latent shape	Subspace of $\mathbb{R}^{D_{\text{ambient}}}$. The geometric image of $\mathcal{X}$ under E .
$\mathbb{R}^{D_{\text{ambient}}}$	The latent ambient	Euclidean container; dimension fixed by the architecture’s exposed
D_{ambient}	Latent ambient dimension	Positive integer. The quantity Theorem 4.8 bounds from below.
$\mathcal{H}$	The function space	Set of maps $\mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ realisable by the architecture. Mentioned for completeness; not load-bearing in this paper.
θ	A parameter realisation	Finite weight configuration selecting one $E \in \mathcal{H}$. Not load-bearing in this paper.

Quotient atlas data. A *quotient atlas* on $\mathcal{X}$ is a tuple specifying how $\mathcal{X}$ decomposes over a base into chamber-indexed fibre objects with per-chamber faithfulness contracts (Def. 3.1).

Symbol	Concept	Type and notes
$\mathcal{Q}$	A quotient atlas	The full tuple $(B, \pi_B, A, \{B_\alpha\}, \{Q_b\}, \{(C_\alpha, \mathcal{T}_\alpha)\})$.
B	The base space	Continuous topological space (typically a bounded subset of $\mathbb{R}^k$, or a compact stratified space) over which fibre objects vary.
π_B	The base map	Continuous $\pi_B : \mathcal{X} \setminus N \rightarrow B$.
b	A base point	$b \in B$.
A	Regime label set	Finite set.
α	A regime label	$\alpha \in A$.
B_α	Chamber	Open subregion of B corresponding to regime α . The chambers $\{B_\alpha\}$ form a decomposition of B (partition or overlap, Defs. 3.2–3.3).
R_α	Region of $\mathcal{X}$	$R_\alpha = \pi_B^{-1}(B_\alpha)$. The portion of $\mathcal{X}$ whose base coordinate lies in chamber α .
Q_b	Fibre object at b	Either the fibre $\pi_B^{-1}(b)$ itself, or a quotient of it specified per chamber. The fibre type is constant on each chamber.
Q_α	Chamber-common fibre type	The homeomorphism class shared by all Q_b for $b \in B_\alpha$. Standard quotient-symbol convention; in the atlas setting, Q is parameterised over B and labelled by α . The single-quotient case of Papers I–III is $ A = 1$.
π_α	Per-region projection	$\pi_\alpha : R_\alpha \twoheadrightarrow Q_\alpha$. Auxiliary; factors through π_B in the partition case.
C_α	Faithfulness contract	Per-chamber structure-preservation requirement (topological, bi-Lipschitz, smooth, isometric, ultrametric, or structure-faithful; see Remark 3.5).
$\mathcal{T}_\alpha$	Target class	Per-chamber class of spaces into which Q_α may be C_α -faithfully embedded.
W_b	A specific target	$W_b \in \mathcal{T}_\alpha$ realising the embedding of Q_b .
N	Negligible set	Closed subset of $\mathcal{X}$, excluded from the atlas (boundaries, critical sets). Scope per the contract’s dimension notion.

Regime-assignment maps.

Symbol	Concept	Type and notes
ι	Regime-assignment function (partition)	$\iota : \mathcal{X} \setminus N \rightarrow A$. Single-valued; continuous on chamber interiors, discontinuous across chamber boundaries.
Λ	Regime-assignment multifunction (overlap)	$\Lambda : \mathcal{X} \setminus N \rightarrow 2^A$. Sends x to the set of chambers containing $\pi_B(x)$ (Def. 3.3).
$\mathcal{N}(\{B_\alpha\})$	Nerve of chamber cover	Simplicial complex encoding chamber overlap structure.

Dimension invariants. The floor theorem relates several distinct dimension quantities; the table makes the distinctions explicit.

Symbol	Concept	Type and notes
d_B	Base embedding dimension	$d_B = \text{emb}_{\text{top}}(B)$ — minimum d such that B admits a topological embedding into $\mathbb{R}^d$. Distinct from covering dimension; e.g., $\dim_{\text{cov}}(S^1) = 1$ but $\text{emb}_{\text{top}}(S^1) = 2$. (Def. 3.12.)
ι_B	Base embedding	$\iota_B : B \hookrightarrow \mathbb{R}^{d_B}$ realising the embedding.
$\text{emb}(Q_\alpha)$	Per-chamber topological embedding dimension	Minimum d with $Q_\alpha \hookrightarrow \mathbb{R}^d$. Per-region cost.
$\dim_{\text{cov}}$	Covering dimension	Standard topological covering dimension. Distinct from emb_{top} in general.
$\dim$	Dimension of a Euclidean factor	Plain integer dimension of $L_B, L_F, \mathbb{R}^{D_{\text{ambient}}}$.

Encoder structure.

Symbol	Concept	Type and notes
ρ	Region-recognition function	$\rho : E(\mathcal{X} \setminus N) \rightarrow A$. Defined on the latent shape, not on all of $\mathbb{R}^{D_{\text{ambient}}}$ (Def. 4.3).
$\tilde{E}_\alpha$	Per-chamber encoder factor	$\tilde{E}_\alpha : Q_\alpha \rightarrow \mathbb{R}^{D_{\text{ambient}}}$. The factorisation of $E _{R_\alpha}$ through π_α .
L_B	Base-coordinate latent factor	Linear subspace of $\mathbb{R}^{D_{\text{ambient}}}$ in the base-split case (Def. 4.4); Theorem 4.8 implies $\dim L_B \geq d_B$.
L_F	Fibre-coordinate latent factor	Linear complement, $\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$; Theorem 4.8 implies $\dim L_F \geq \sup_b \text{emb}(Q_b)$.

2 Two faces of atlas structure

Two motivating examples, chosen to occupy different positions in the space of atlas structures. The first is simpler; the second is where the atlas framing was forced. Both are worked fully in Paper VII; we extract only what motivates the formalism.

2.1 Color Fantasy: continuous base with physically-constrained regions

The Color Fantasy system (Paper VII §8) is a ship-hotel HVAC setting with four thermal control regimes separated by hard threshold switches on internal-temperature setpoints under ambient heat-loss conditions. The regime boundaries are joint thresholds on two continuous physical variables: the cabin internal temperature T_{cabin} and the car-bay temperature T_{bay} . The base is therefore two-dimensional:

$$B \subseteq \mathbb{R}^2, \quad (T_{\text{cabin}}, T_{\text{bay}}) \in B, \quad d_B = 2.$$

The four regimes correspond to open regions of B :

- *Cooling*. Cabin temperature above the cooling setpoint; air conditioning active; heating inactive.
- *Neutral*. Cabin temperature between the cooling and heating setpoints; neither heating nor cooling primary.
- *Moderate heating*. Cabin temperature below the heating setpoint but above the high-intensity threshold.

- *Strong heating.* Cabin temperature below the high-intensity threshold; full heating load.

Physics constrains B to a proper subset of $\mathbb{R}^2$: the cabin and car bay are thermally coupled through the ship structure, so combinations such as ($T_{\text{cabin}} > 24^\circ\text{C}$, $T_{\text{bay}} < 5^\circ\text{C}$) cannot be realised in steady state. We *assume* the realised cabin–bay state envelope contains a two-dimensional region with nonempty interior in the measured coordinates; under this assumption $d_B = 2$, and the data of Farkas [2026g] support the assumption (both temperatures vary substantially within and across regimes). The contrary case — a genuinely tight operational coupling in which one coordinate is a deterministic function of the other on the realised data — would reduce the effective base dimension and is not observed, but is the right caveat to flag.

Three structural features follow.

The base is continuous, not discrete. Earlier drafts of this paper, and the author’s own initial intuitive atlas declaration for Color Fantasy in Paper VII (which predated explicit awareness of atlas formalism), treated the four regimes as a finite discrete index set. That reading conflated qualitative regime labels with the underlying continuous physical variables driving the switches. The correct reading is that the four regime labels name qualitative local types of a continuous two-dimensional base — structurally parallel to the double pendulum (§2.2), which has a continuous one-dimensional base (energy) partitioned into four qualitative regime types.

Shared content is cyclic but sits in the fibre. The available Color Fantasy data covers a single year, so the annual phase $\varphi_{\text{year}} \in S^1$ is not distinguishable in the data and does not appear in the atlas. A day–night cyclic variable $\varphi_{\text{day}} \in S^1$ is present in the data and is regime-independent: it has the same interpretation in all four regimes. Under the present modelling choice it belongs in each regime’s local quotient Q_α rather than in the base; an alternative modelling choice would put it in the base, raising d_B and correspondingly lowering $\text{emb}(Q_\alpha)$. A multi-year extension would add φ_{year} as a third base coordinate under either choice.

Within each regime, additional regime-specific content. Each regime α carries its own delay structure (thermal-mass memory of the hull, captured by the delay variable in Paper VII §8’s GCD architecture), its own operative subset of control variables, and its own regression relationship between inputs and energy consumption. These contribute additional dimensions to $\text{emb}(Q_\alpha)$ beyond the shared S^1 day–night factor.

2.2 Double pendulum: continuous-base atlas

The double pendulum (Paper VII §7) is a different phenomenon. The configuration space is $Q = T^2$; the full phase space is $\mathcal{X} = T^*T^2 \cong T^2 \times \mathbb{R}^2$. The Hamiltonian H is a smooth function of state with four non-degenerate critical points: the stable down-down equilibrium at $E = -3g$ (Morse index 0), two saddle equilibria at $E = -g$ and $E = +g$ (index 1), and the upside-down-upside-down equilibrium at $E = +3g$ (index 2). By Morse theory on the sublevel sets [Milnor, 1963b], [Smale, 1970b], the energy surface $\Sigma_E = H^{-1}(E)$ undergoes topology change at each critical energy via handle attachment. The regular-level 3-manifolds are:

Energy range	Σ_E
$E < -3g$	$\emptyset$
$-3g < E < -g$	S^3
$-g < E < +g$	$S^2 \times S^1$
$+g < E < +3g$	$\#_2(S^2 \times S^1)$
$E > +3g$	T^3

The final T^3 topology is supported by the high-energy asymptotic argument: for $E \gg +3g$ the potential is negligible and Σ_E is a trivial ellipse bundle over T^2 , hence $T^2 \times S^1 = T^3$ (Paper VII, §*dp_atlas*).

Each energy regime has its own 3-manifold topology and its own effective dynamical character. The low-energy regime is approximately integrable (KAM tori near the stable equilibrium [Arnold, 1989, Lichtenberg and Lieberman, 1992]); the intermediate regimes are chaotic without global action-angle structure; the high-energy regime admits an approximate $U(1) \times U(1)$ action-angle structure on a different topological base.

Four features distinguish this from Color Fantasy.

The base is continuous. Energy E is a continuous function of state; regime boundaries are smooth hypersurfaces in $\mathcal{X}$, not discrete thresholds.

Local quotient structure changes smoothly within regimes and discontinuously across them. Within a regime the fibre topology is fixed; at Morse-critical values it jumps.

Regime boundaries are themselves nontrivial. The separatrices at $E = \pm g$ are measure-zero submanifolds with their own structure (homoclinic and heteroclinic connections associated with the index-1 saddles). Further, the stable down-down equilibrium at $E = -3g$ lies on the boundary of the low-energy S^3 regime in the atlas-partition sense, yet is dynamically central to that regime — most training data in Paper VII concentrates near it. Boundary sets of the atlas partition can contain points of central dynamical importance.

Dynamical topology cannot be recovered by global diagnostics. Applying persistent homology to the full trajectory data yields the homotopy type of whatever region of phase space is sampled, which retracts onto T^2 regardless of regime (Paper VII, data-adequacy discussion). The informative diagnostic is *region-stratified* persistent homology — per-energy-bin barcodes compared against the regional topology of each Morse region — but this requires regime boundaries to be identified first. §6 returns to this.

2.3 What the two examples establish

Both Color Fantasy and the double pendulum are continuous-base atlases with four qualitative regime types partitioning the base. They differ on three axes that matter for the formalism.

Base dimension. The double pendulum has $d_B = 1$ (a bounded energy interval). Color Fantasy has $d_B = 2$ (the joint cabin-bay temperature space, physically constrained to a subset of $\mathbb{R}^2$).

Memoryless vs. memory. The double pendulum is memoryless: Hamiltonian dynamics are time-reversible, there is no hysteresis, and the memoryless compatibility condition formalised as Assumption 4.2 is satisfied. Color Fantasy is not: thermal-mass memory of the ship hull is a real physical effect and is modelled explicitly by the delay variable in Paper VII §8’s GCD architecture. Theorem 4.8, as stated, covers the memoryless

approximation of Color Fantasy — Example 4.12 applies in that approximation, where the delay is treated as regime-local regression input rather than as overlap compatibility data. The full memory structure is flagged as an open problem (§7.2.1). This is a scope limitation of the present theorem, not of the atlas formalism, which accommodates overlap structure with memory through the compatibility-strength axis of §3.3 below.

Regime-transition regularity. Double pendulum regime transitions are smooth Morse handle attachments at critical energy values — the classical local models of §3.4 apply. Color Fantasy regime transitions are hard threshold crossings in the base variables with no smooth handle structure between adjacent regimes; the per-regime local quotient structure simply switches across regime boundaries.

The Color Fantasy atlas declaration in Paper VII was constructed intuitively by the present author without explicit awareness of atlas formalism at the time. The reframing above — continuous two-dimensional base, regime labels as qualitative colouring, day–night S^1 in the fibre, thermal memory acknowledged as out-of-scope for Theorem 4.8 — is the formalism catching up with what the empirical work had already done.

The formalism should be a framework with axes of variation, not a single definition that forces all atlases into one mould. §3 introduces the formal machinery and the *compatibility-strength axis* along which these two examples sit at different points.

Neither Color Fantasy nor the double pendulum has been independently verified as the uniquely correct atlas structure for its underlying problem (Paper VII §8 explicitly flags the Color Fantasy decomposition as a modelling choice). For realistic problems the atlas structure is a hypothesis; part of §5–6’s task is to describe what would count as evidence for or against a given atlas hypothesis.

3 Formalism

3.1 Quotient atlas

Definition 3.1 (Quotient atlas). Let $\mathcal{X}$ be a compact Hausdorff topological space — typically a manifold, an atlas of manifolds, or a stratified or singular space. A *quotient atlas* on $\mathcal{X}$ is a tuple

$$\mathcal{Q} = (B, \pi_B, A, \{B_\alpha\}_{\alpha \in A}, \{Q_b\}_{b \in B}, \{(C_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in A})$$

where:

- (1) B is the *base space*: a topological space (typically a bounded subset of $\mathbb{R}^k$, or a compact stratified space) over which fibre objects are indexed.
- (2) $\pi_B : \mathcal{X} \setminus N \rightarrow B$ is the *base map*, continuous, where $N \subseteq \mathcal{X}$ is closed and negligible with respect to the dimension notion fixed by the chosen contracts (see Remark 3.5).
- (3) A is a finite set of *regime labels*, and $\{B_\alpha\}_{\alpha \in A}$ is a decomposition of B into qualitative *chambers*: subregions of B whose specific topological character (open and pairwise disjoint, open and overlapping, or strata of varying dimension) is fixed by the chamber-meeting case in Def. 3.2, Def. 3.3, or Def. 3.4 below. The associated *regions* of $\mathcal{X}$ are $R_\alpha = \pi_B^{-1}(B_\alpha)$.
- (4) For each $b \in B$, Q_b is the *fibre object* at b : either the fibre $\pi_B^{-1}(b)$ itself, or a quotient of it specified per chamber. The fibre type is constant on each chamber: for $b \in B_\alpha$, the Q_b are mutually homeomorphic and we write Q_α for the common

type. Chamber-wise constancy of fibre type is a modelling commitment of the present formalism, not a property of all atlas-structured systems: real systems may exhibit smoothly varying fibre topology within a chamber, or bifurcations not aligned with chamber boundaries. Such systems require a finer chamber decomposition or a more general formalism; they are listed as an open direction (§7).

- (5) For each chamber α , $(C_\alpha, \mathcal{T}_\alpha)$ is the *faithfulness contract* and *target class* for that chamber: each Q_b for $b \in B_\alpha$ admits a C_α -faithful embedding into some target $W_b \in \mathcal{T}_\alpha$.

Informal preview (made precise in Def. 4.4). The atlas may admit a *global product structure* in which a continuous selection of fibre embeddings exists making the assembled map $x \mapsto (\pi_B(x), Q_{\pi_B(x)})$ -coordinate continuous on $\mathcal{X} \setminus N$; this is the case the floor theorem of §4 addresses. Weaker compatibility levels are admitted and discussed in §3.3.

Quotient atlases split into three types according to how the chambers meet in B : pairwise disjoint (partition, Def. 3.2), pairwise overlapping on open subregions (open overlap, Def. 3.3), or meeting on lower-dimensional strata that themselves carry fibre structure (stratified meet, Def. 3.4). The floor theorem of §4 is proved for the partition case; the other two are open (§7.2.2, §7.2.3).

Definition 3.2 (Partition atlas). A quotient atlas $\mathcal{Q}$ is a *partition atlas* if the chambers $\{B_\alpha\}_{\alpha \in A}$ are open subregions of B , pairwise disjoint modulo a negligible boundary set: $B_\alpha \cap B_\beta = \emptyset$ for $\alpha \neq \beta$, and consequently $R_\alpha \cap R_\beta = \emptyset$. The *regime-assignment function*

$$\iota : \mathcal{X} \setminus N \rightarrow A, \quad \iota(x) = \alpha \text{ when } \pi_B(x) \in B_\alpha,$$

is well-defined and single-valued. Note that ι is the composition of the continuous base map π_B with the (typically discontinuous) chamber classifier $B \rightarrow A$; ι is continuous on the interior of each chamber but switches discontinuously across chamber boundaries.

Definition 3.3 (Overlap atlas). A quotient atlas $\mathcal{Q}$ is an *overlap atlas* if the chambers $\{B_\alpha\}_{\alpha \in A}$ are open subregions of B and at least one pair B_α, B_β ($\alpha \neq \beta$) has nonempty intersection. The regime-assignment data is then the multifunction

$$\Lambda : \mathcal{X} \setminus N \rightarrow 2^A, \quad \Lambda(x) = \{\alpha \in A : \pi_B(x) \in B_\alpha\},$$

equivalently encoded by the nerve $\mathcal{N}(\{B_\alpha\})$ of the chamber cover of B .

Definition 3.4 (Stratified-meet atlas). A quotient atlas $\mathcal{Q}$ is a *stratified-meet atlas* if the chambers $\{B_\alpha\}_{\alpha \in A}$ form a topological stratification of B : chambers are pairwise disjoint, of possibly varying dimensions, and the closure of each chamber satisfies the frontier condition

$$\overline{B_\alpha} \subseteq B_\alpha \cup \bigcup \{B_\beta : \dim B_\beta < \dim B_\alpha\}.$$

Each chamber, of whatever dimension, carries its own fibre type Q_α as supplied by the atlas data. The regime-assignment function $\iota : \mathcal{X} \setminus N \rightarrow A$ is single-valued: each base point lies in exactly one chamber.

The figure-eight merge is the canonical example. Take B to be a tripod (two arms entering a junction, one arm exiting), with two top-dimensional chambers over the entering arms (each carrying fibre type S^1), one top-dimensional chamber over the exiting arm (carrying its own fibre type), and one zero-dimensional chamber at the junction point (carrying fibre type $S^1 \vee S^1$, the wedge of the two adjacent S^1 's). The junction fibre is not the intersection or union of the adjacent fibres but a wedge sum, and the relation between the junction fibre and the adjacent fibres is part of the atlas data. The compatibility-strength axis (§3.3) is currently formulated for partition and open-overlap atlases; the analogous hierarchy for stratified-meet atlases — frontier-compatibility levels relating Q_α on a lower-dimensional chamber to the Q_β on adjacent higher-dimensional chambers — is named as a separate open problem (§7.2.3).

Remark 3.5 (Canonical contract–target pairs and the branched hierarchy). A *faithfulness contract* C is a notion of structure-preservation on maps $Q_\alpha \rightarrow T_\alpha$ drawn from an established hierarchy of such notions. A *target class* $\mathcal{T}$ is a class of spaces serving as embedding targets for C . The known contracts do not form a single linear chain: they fall into several branches that are pairwise incomparable at the level of the maps they admit. The canonical $(C, \mathcal{T})$ pairs include:

Archimedean (Euclidean-target) family. These contracts form a linear refinement chain among themselves.

- *Topological:* Q_α admits a topological embedding into $\mathbb{R}^{m_\alpha}$ for some finite m_α . Target class $\mathcal{T} = \{\mathbb{R}^N : N \in \mathbb{N}\}$. Dimension invariant: covering dimension (when integer-valued). This is the contract of Paper III §2.3 and the case in which Theorem 4.8 below is proved.
- *Bi-Lipschitz with distortion bound D :* Q_α admits a bi-Lipschitz embedding into a doubling metric space (Paper III §2.4, Proposition 2.4; Naor and Neiman 2012). Target class $\mathcal{T}$ = doubling metric spaces or $\mathbb{R}^N$ with snowflake-distorted metric. Dimension invariant: Assouad dimension with parametric dependence on D .
- *Smooth (C^k , $k \geq 1$):* Q_α is a smooth manifold and the embedding is C^k (Whitney, Haefliger). Target class $\mathcal{T} = \{\mathbb{R}^N\}$ with smooth structure. Dimension invariant: smooth-manifold dimension with Whitney or Haefliger bounds.
- *Isometric (C^1 or C^∞):* Nash–Kuiper or Nash–Günther isometric embedding into $\mathbb{R}^N$ (Paper III §2.4, Theorem 2.4). Dimension invariant: isometric embedding dimension, which can exceed the topological floor substantially.

Non-Archimedean (ultrametric-target) family. A parallel branch, not a refinement of the Archimedean one.

- *Ultrametric:* Q_α carries an ultrametric structure and admits an ultrametric-faithful embedding into $\mathbb{Q}_p^{m_\alpha}$ or into a general ultrametric target, following exploratory foundations in Martins [2026] and N’guessan et al. [2025]. The Archimedean and ultrametric contracts are rigorously incomparable: a metric space is bi-Lipschitz embeddable into an ultrametric space iff it is uniformly disconnected, a property no connected Euclidean space possesses, while compact ultrametrics of infinite Assouad dimension admit no bi-Lipschitz embedding into any finite-dimensional Euclidean space. The Ishiki–Schikhof non-Archimedean Arens–Eells theorem realizes every ultrametric space isometrically into some (generally

infinite-dimensional) valued-field extension; no finite-dimensional $\mathbb{Q}_p^d$ embedding-floor theorem analogous to Theorem 4.8 is currently established. See §7 for the associated open problem.

Structure-faithful (physics-type) family. A further branch, incomparable with the metric families above.

- *Symplectic, Hamiltonian-conjugacy, Lagrangian, equivariant, measure-preserving, hard-constrained operator:* Q_α carries a geometric or algebraic structure (symplectic form, conserved quantity, group action, invariant manifold) and the embedding exactly preserves it. These contracts are strictly stronger than topological but are generally incomparable with bi-Lipschitz and isometric contracts: a linear symplectic map $\text{diag}(\lambda, \lambda^{-1})$ has unbounded condition number, so symplecticity does not imply a uniform bi-Lipschitz bound; conversely, a bi-Lipschitz map need not preserve any of these structures. Quantitative bridges (e.g. non-squeezing under a Lipschitz cap, Naor and Neiman 2012-style distortion estimates) give hybrid contracts rather than a linear chain. No atlas-level floor theorem is established for this family; see §7.

The homogeneous case. When $(C_\alpha, \mathcal{T}_\alpha) = (C, \mathcal{T})$ for all $\alpha \in A$, the atlas is said to be *homogeneous* with contract $(C, \mathcal{T})$. The homogeneous-topological-Euclidean case is the scope of the proved Theorem 4.8 (§4). Homogeneous atlases at the other Archimedean contracts, at the ultrametric contract, and at the structure-faithful contracts are open problems (§7). Heterogeneous atlases, in which distinct regions carry distinct contracts, are admitted by Definition 3.1 but require a cross-species comparison calculus on overlaps that is not developed here; they too are listed as an open problem (§7).

Remark 3.6 (Positioning relative to existing formalisms). A quotient atlas has close relatives in several mathematical frameworks, none of which is a perfect match. *Abstract stratified sets* in the Whitney–Mather tradition [Whitney, 1965], [Mather, 1970], [Goresky and MacPherson, 1988], [Pflaum, 2001] handle partitions into smooth strata with compatible tubular-neighbourhood control data, and support variable local dimension and Morse-type transitions natively; the quotient atlas is closest to this framework in spirit, though Paper IV does not require condition-(B) regularity at this level of generality. *Topological stacks and Lie groupoids* [Noohi, 2005], [Metzler, 2003], [Moerdijk and Pronk, 1997] are the right formal upgrade at the strongest compatibility level only, where overlap data is cocycle-coherent; the atlas is then a presentation of a stack. *Orbifolds* [Moerdijk and Mrčun, 2003], [Lerman, 2010] are a special case: locally finite-group quotients with discrete isotropy, narrower than “local quotient varies by regime.” *Differential spaces* [Śniatycki, 2013], [Cushman and Śniatycki] provide a smooth-function framework for singular quotients, more general on singular quotient structure but weaker on explicit overlap-compatibility data. The quotient atlas is closest to an abstract stratified set enriched by a region-indexing base and per-region quotient data, with compatibility strength a separate axis (§3.3).

Remark 3.7 (Reduction to single quotient). A homogeneous partition atlas with $|A| = 1$ and $R_\alpha = \mathcal{X} \setminus N$ for N empty or of measure zero recovers the single-quotient setting of Papers I–III at the corresponding contract. The present theory is a strict generalisation.

Remark 3.8 (Distributional reading). For any Borel probability measure μ on $\mathcal{X}$ with $\mu(N) = 0$ — which holds for any μ absolutely continuous with respect to a measure that vanishes on lower-dimensional subsets — the atlas covers μ -almost every point. Empirical statements in §5–§6 invoke this reading. Structural theorems in §3–§4 do not require a distribution.

Remark 3.9 (Structure on the excluded set). The excluded set $N = \mathcal{X} \setminus \bigcup_{\alpha} R_{\alpha}$ is omitted from the partition condition but not from the underlying space $\mathcal{X}$. Points of N are limits of points in adjacent regions; their neighbourhoods are covered by adjacent regions’ local quotients. N may carry structure of independent interest — fixed points of a dynamical flow, separatrices, threshold hypersurfaces with transition dynamics, homoclinic and heteroclinic connections associated with saddles. A fixed point lying in N can be dynamically central to an adjacent region even though it lies on the region’s boundary in the atlas sense; a saddle’s stable and unstable manifolds organise the transitions between adjacent regions. §4–§6 treat N as atlas-level negligible; structure within N is an independent object of study (§7). All atlas-level statements — in particular the base map π_B , the regime-assignment function ι , the multifunction Λ , and the proof of Theorem 4.8 — are understood as holding on $\mathcal{X} \setminus N$. Continuity and faithfulness conditions on the encoder E are similarly stated up to restriction to $\mathcal{X} \setminus N$; behaviour on N itself is unconstrained by the atlas data.

Remark 3.10 (Ambiguity of the atlas). For a given $\mathcal{X}$ with its data-generating distribution, the quotient atlas is not uniquely determined. The same underlying problem may admit multiple atlases — finer and coarser partitions, or partitions along different axes. This is structurally different from the single-quotient case: the degrees of freedom are strictly richer (choice of partition vs. overlap structure, choice of base, choice of contract per region, and per-region relations, rather than just an equivalence relation on a single space). We return to atlas selection in §5–§6.

Remark 3.11 (Generator-specified quotient targets). When $Q_{\alpha} = \text{Gen}(\mathfrak{g}_{\alpha})$ for a generator specification $\mathfrak{g}_{\alpha}$ in the sense of Paper VI [Farkas, 2026f], the faithfulness contract C_{α} and target class $\mathcal{T}_{\alpha}$ are typically inherited from the generator class: a Lie-symmetry generator with free proper action specifies a homogeneous-space target with natural Riemannian structure (Euclidean-target, smooth or isometric contract); a discrete-symbolic or taxonomic generator specifies an ultrametric target (ultrametric contract); a Hamiltonian or symplectic generator specifies a symplectic target (structure-faithful contract); a regression-model generator specifies a Euclidean target with the contract set by the model’s regularity class. The generator framework thus supplies both the quotient and its natural contract–target pair, reducing the atlas specification to the partition and the generator family. Inheritance is not automatic for generator classes mixing Lie-type and non-Lie-type content (Remark 5.3 below); such cases require the contract to be declared separately.

3.2 Base dimension

Definition 3.12 (Base embedding dimension). The *base embedding dimension* d_B of the atlas is the topological-embedding dimension of the base space B :

$$d_B := \text{emb}_{\text{top}}(B) = \min\{d \in \mathbb{N} : B \hookrightarrow \mathbb{R}^d \text{ topologically}\}.$$

A specific embedding realising this minimum is denoted $\iota_B : B \hookrightarrow \mathbb{R}^{d_B}$.

The base embedding dimension is in general distinct from the topological covering dimension $\dim_{\text{cov}}(B)$. They coincide for many cases of interest (bounded subsets of $\mathbb{R}^k$ that are themselves embeddable in $\mathbb{R}^k$), but diverge in general: $\dim_{\text{cov}}(S^1) = 1$ while $\text{emb}_{\text{top}}(S^1) = 2$, since S^1 does not embed in $\mathbb{R}^1$. For discrete B (the degenerate case $|A|$ finite with B a finite point set), $d_B = 0$.

This definition presumes that B is topologically embeddable in some finite-dimensional Euclidean space. This holds for both motivating examples (continuous $B \subseteq \mathbb{R}$ for the double pendulum energy axis, continuous $B \subseteq \mathbb{R}^2$ for Color Fantasy) and for atlases whose base is a manifold or CW complex. The general case is addressed by the scope restriction in §4.1.

3.3 The compatibility-strength axis

An atlas without overlap structure is an atomised collection of local quotient problems. What distinguishes a quotient atlas from a disjoint union of independent problems is a compatibility condition on how π_α and π_β relate where regions meet. The strength of this condition is the principal design axis of the theory, and maps onto the standard descent-strength hierarchy of sheaf and stack theory [Vistoli, 2005], [Noohi, 2005].

We distinguish four levels, from weakest to strongest. For brevity, when $|A| \geq 2$, denote overlaps by $U_{\alpha\beta} := R_\alpha \cap R_\beta$; the discussion extends to higher-order intersections naturally.

(a) Disjoint. The regions meet only on the excluded set: $U_{\alpha\beta} \subseteq N$ for all $\alpha \neq \beta$. No compatibility condition is imposed; local quotients are independent. The atlas is a labelled partition. *Color Fantasy is at level (a).* Regimes meet only on threshold hypersurfaces of measure zero; there is no nontrivial sense in which the cooling-regime quotient “agrees with” the heating-regime quotient on an overlap.

(b) Set-theoretic compatibility. Regions may genuinely overlap: $U_{\alpha\beta}$ has nonempty interior. Compatibility at the level of the induced equivalence: for $x, x' \in U_{\alpha\beta}$,

$$\pi_\alpha(x) = \pi_\alpha(x') \iff \pi_\beta(x) = \pi_\beta(x').$$

The local quotients partition the overlap into the same fibres; they may name those fibres differently but agree on which points are identified.

(c) Isomorphism on overlap. Stronger than (b). For each pair α, β with nonempty overlap, there exists a homeomorphism (or diffeomorphism, depending on the contract) $\phi_{\alpha\beta} : \pi_\alpha(U_{\alpha\beta}) \rightarrow \pi_\beta(U_{\alpha\beta})$ such that $\pi_\beta|_{U_{\alpha\beta}} = \phi_{\alpha\beta} \circ \pi_\alpha|_{U_{\alpha\beta}}$. The two local quotients are identified as quotient objects on the overlap, not merely inducing the same equivalence.

(d) Cocycle (effective descent). Level (c) plus the triple-overlap condition: for $U_{\alpha\beta\gamma}$ nonempty, $\phi_{\beta\gamma} \circ \phi_{\alpha\beta} = \phi_{\alpha\gamma}$ on $\pi_\alpha(U_{\alpha\beta\gamma})$. This is the standard sheaf-theoretic descent condition and, in stack-theoretic language, the point at which the atlas assembles into a coherent global object. The colimit of the $\{\pi_\alpha\}$ is well-defined up to canonical isomorphism, and the atlas is a chart decomposition of a single coherent global quotient [Noohi, 2005].

The double pendulum can be interpreted as level (c) for pairs of adjacent regimes under a standard smooth-structure interpretation, though Paper IV does not develop the full

cocycle theory for it. Pragmatically, per-regime quotients are compatible on their one-sided boundaries via the Morse-cobordism local model (§3.4) rather than via genuine open-set overlaps; the level-(c) interpretation is a structural reading, not a derivation.

Remark 3.13 (Why we do not fix a single level). Real problems sit at different levels. Color Fantasy-type atlases (hard regime switches, no overlap dynamics) genuinely cannot support conditions beyond (a). Smooth Morse-stratified atlases with overlap bands can support (c). Globally coherent atlases arising from a single smoothly varying structural parameter can satisfy (d). Paper IV therefore develops the theory as a framework indexed by compatibility level, not a theorem for a single universal definition. §4 proves a floor theorem for partition atlases (level (a)) at the homogeneous topological-Euclidean contract; floor theorems for atlases at levels (b)–(d), with overlap gluing data contributing to the bound, are open (§7.2.2). §5 discusses selection over families at heterogeneous compatibility levels; §6 discusses emergent discovery, which requires at least (b) for faithful-summary diagnostics to have content.

Remark 3.14 (What the levels buy). Level (a) suffices for per-region analysis but has no global structure beyond the partition. Level (b) permits coherent region-recognition but not canonical identification of objects across charts. Level (c) permits per-pair identification but without guaranteed consistency across triples. Level (d) is the only level at which the atlas is a coherent global mathematical object in the classical sense. Stronger is not always better empirically, because (c) and (d) commit to identification data that real problems may not genuinely possess.

Remark 3.15 (Compatibility strength is not regularity). The compatibility-strength axis is orthogonal to the Whitney regularity conditions (A), (B) for stratified sets [Whitney, 1965], [Mather, 1970]. Condition-(A)/(B) regularity is a *geometric* condition on how strata meet — governing local cone-bundle triviality and tubular-neighbourhood control — whereas compatibility strength is a *structural* condition on how local quotients identify on overlaps. We do not attempt to combine the two into a single hierarchy.

Remark 3.16 (Scope: partition and open overlap only). The four levels (a)–(d) above are formulated for chamber-meeting cases in which overlaps, when present, are open subregions of B — that is, for partition atlases (Def. 3.2) and overlap atlases (Def. 3.3). The analogous hierarchy for stratified-meet atlases (Def. 3.4) would replace “isomorphism on the open overlap” with *frontier-compatibility* conditions relating the fibre Q_α on a lower-dimensional chamber to the fibres Q_β on adjacent higher-dimensional chambers — via wedge sums, pushouts, or other gluing prescriptions specified by the atlas data. A worked development of frontier-compatibility levels is not given here. The partition floor theorem of §4 should be read as isolating the base-plus-fibre mechanism in the simplest chamber-meeting case; overlap and stratified-meet floor theorems require additional compatibility data and are open problems (§7.2.2, §7.2.3).

3.4 Local models of atlas transitions

The excluded set N is not a blank object. §2 and Remark 3.9 noted that N may contain structured geometric data. We now record the local models that drive §4’s compatibility-cost calculation.

Regular transitions (within a regime). For x in the interior of some R_α , the local quotient π_α is defined on a neighbourhood and behaves continuously there. No special machinery is required.

Morse transitions (smooth-base atlases). For atlases whose regions are level-set families of a smooth function $f : \mathcal{X} \rightarrow \mathbb{R}$ with non-degenerate critical points, the transition between adjacent regions at a critical value c of index k admits Morse’s lemma: in suitable local coordinates near a critical point p with $f(p) = c$ and index k ,

$$f = c - (x_1^2 + \cdots + x_k^2) + (x_{k+1}^2 + \cdots + x_n^2).$$

The sublevel set $\{f \leq c\}$ is obtained from $\{f \leq c - \epsilon\}$ by attaching a k -handle. The regular-level set $\{f = c + \epsilon\}$ is obtained from $\{f = c - \epsilon\}$ by a surgery whose local model is determined by the index k and the ambient dimension [Milnor, 1963b]. Stratified Morse theory [Goresky and MacPherson, 1988] treats analogous critical-value transitions on stratified spaces with tangential-normal decompositions of local Morse data.

Hard-threshold transitions. For atlases whose regions are separated by threshold hypersurfaces in the base with no smooth interpolation between adjacent regions (Color Fantasy type, where the base is continuous but the local-quotient data switches discontinuously across threshold crossings), no local Morse model applies. The transition is a discontinuity in the local quotient data: fibre structure, dimension, active variables, or quotient type can change arbitrarily across a threshold. Within each region the local quotient is as regular as assumed; between regions, no deformation connects them. (A genuinely discrete-base atlas, with the base A itself a finite discrete set, is the degenerate $d_B = 0$ case; it is admitted by Def. 3.1 but is not the structure of the Color Fantasy example.)

Remark 3.17 (Mixed atlases). Nothing prevents atlases with both smooth Morse transitions and discrete transitions. A physical system with continuous energy stratification within a dynamical mode but discrete mode switches between modes is a natural example. §4’s floor theorem applies to such mixed atlases directly when the encoder is base-split and transitions are regular; singular transitions of either kind contribute excess dimension beyond the clean two-term floor, as catalogued in Remark 4.9 and §7.3.1.

3.5 Categorical reading (optional)

The formalism of §3.1–§3.4 can be read through a standard categorical lens. The content of this subsection is interpretive; no result in §4–§6 depends on it. Readers uninterested in the categorical view may skip it without loss.

Partition atlas as coproduct. For a partition atlas (Def. 3.2), the decomposition $\mathcal{X} \setminus N = \coprod_{\alpha \in A} R_\alpha$ is literally a coproduct in the category **Top** of topological spaces (or in the category of compact Hausdorff spaces restricted to $\mathcal{X} \setminus N$). The atlas data $\{\pi_\alpha : R_\alpha \rightarrow Q_\alpha\}$ is then a functor from the discrete category indexed by A into **Top**.

Overlap atlas as colimit over the nerve. For an overlap atlas (Def. 3.3), let $\mathcal{N}(\{R_\alpha\})$ denote the nerve category: objects are nonempty intersections $U_{\alpha_0} \cap \cdots \cap U_{\alpha_k}$; morphisms are inclusions. The atlas data is a functor $\mathcal{N}(\{R_\alpha\}) \rightarrow \mathbf{Top}$, and $\mathcal{X} \setminus N$ is realised as the colimit. The partition case is recovered when the nerve is discrete (no nonempty higher intersections).

Compatibility-strength axis as sheaf conditions. The per-region quotient data $\{\pi_\alpha\}$ assembles into a presheaf on the region-indexing poset. Compatibility levels (b), (c), (d) of §3.3 are successively stronger sheaf conditions: level (b) is set-theoretic agreement on overlaps; level (c) is isomorphism of overlap restrictions with specified identifications; level (d) is the cocycle condition, equivalent to effective descent. The extension theorem for sheaves on a base [Vakil, 2024, Ch. 2.5] is the direct machinery behind the claim that compatibility data on the cover determines the global quotient object.

Base-split hypothesis vs. diagonal separatedness. The base-split hypothesis (Def. 4.4) and scheme-theoretic separatedness [Vakil, 2024, Ch. 11] both rule out non-Hausdorff-style gluing pathologies, but they are not equivalent. Scheme-theoretic separatedness is the condition that the diagonal $\Delta : X \rightarrow X \times_Y X$ is a closed immersion — a diagonal-closedness condition that does not, in general, imply a global product structure on the total space. The base-split hypothesis here is strictly stronger: it requires the latent ambient $\mathbb{R}^{D_{\text{ambient}}}$ to factor as a coordinate product $L_B \oplus L_F$ in which L_B recovers the base. The line-with-two-origins counterexample is illustrative for both notions — it is the classical non-separated scheme *and* the encoder pathology where region-recognition data is locally consistent but does not globalise to a coordinate splitting — but the categorical equivalence is not claimed (Remark 4.5).

Per-region images assemble via fibred product. The encoder image $E(\mathcal{X}) \rightarrow A$ has each fibre over $\alpha \in A$ containing the per-region image $\tilde{E}_\alpha(Q_\alpha)$. For partition atlases the global image is the coproduct $\coprod_\alpha \tilde{E}_\alpha(Q_\alpha)$; for overlap atlases satisfying cocycle compatibility it is the colimit of the per-region images along the nerve [Vakil, 2024, Ch. 10]. The base-split hypothesis (Def. 4.4) is what makes this assembly behave as a trivialised product over the base, not merely a fibred product.

Remark 3.18 (Why this is a remark and not the formalism). The body of Paper IV is kept set-theoretic for continuity with Papers I–VI and because the set-theoretic statements admit direct diagnostic use (§5–§6). The categorical reading above is offered as a parallel interpretation that clarifies why certain hypotheses (separation, cocycle compatibility) arise naturally and locates them within established mathematical machinery. Future work on atlas-level gluing, cohomological invariants of atlas structure, stack-theoretic upgrades of the compatibility levels, and cross-species comparison for heterogeneous-contract atlases (§7) would likely be written in the categorical register from the start, drawing on fibred-category and constructible-stack foundations [Vistoli, 2005].

4 The base-split atlas dimension floor

This section states and proves the main structural theorem of the paper: a lower bound on the latent dimension required for an encoder to be quotient-faithful with respect to a given partition atlas under the base-split representation contract. The result extends Proposition 3.1 of Paper III [Farkas, 2026c] from a single global quotient to partition atlases, under the topological contract. Stronger contracts (smooth, bi-Lipschitz, isometric) give strictly stronger floors; the theorem is stated at the topological level and stronger cases are noted as corollaries.

Existing literature suggests the base-plus-fibre shape of the statement, but not this representation-contract lower bound. Natsume’s realisation theorem [Natsume, 1980] shows that every paracompact abstract stratified set of dimension n embeds in $\mathbb{R}^{2n+1}$,

analogously to the classical Menger–Nöbeling bound for metric spaces of covering dimension n ; parametric embedding theorems for n -dimensional maps into a base give total-space bounds with a base-dimension-plus-fibre-dimension flavour [Skopenkov, 2008]; and the Hurewicz dimension-lowering inequality $\dim_G X \leq \dim f + \dim_G Y$ for continuous $f : X \rightarrow Y$ [Hurewicz and Wallman, 1941] provides a fibre-plus-base decomposition for covering dimensions of images. Theorem 4.8 recasts these dimensional heuristics as a lower-bound *contract* — a floor on the latent dimension required for an encoder to be atlas-faithful under the base-split representation hypothesis, not an upper bound on the dimension of a stratified realisation. The encoder-conditional, embedding-dimension framing under the base-split contract is what is new; the inequality itself is close in spirit to Hurewicz (see Appendix B.9). Throughout §4, the input space $\mathcal{X}$ is the compact subset of the underlying physical state space supported by the realised training data, not the full phase space; the theorem’s conclusions apply to that data-supported subset, and extrapolation to unsampled regions of phase space requires separate justification.

The theorem in this section is proved for partition atlases (Def. 3.2) at the homogeneous topological contract. Its proof isolates the base-plus-fibre lower-bound mechanism in the simplest chamber-meeting case. Extending this mechanism to open-overlap atlases requires compatibility of the coordinate splitting with the nerve of the chamber cover; extending it to stratified-meet atlases requires frontier-compatibility data on lower-dimensional chambers. Neither extension is supplied here; both are open problems (§7.2.2, §7.2.3).

4.1 Setup and hypotheses

Assumption 4.1 (Regularity). The results of §4 work in the following setting:

- (1) $\mathcal{X}$ is compact Hausdorff. (For the motivating examples, $\mathcal{X}$ is the compact subset of phase space supported by the realised data — a bounded energy interval for the double pendulum, the realised cabin–bay envelope for Color Fantasy — not the full noncompact phase space.)
- (2) The atlas $\mathcal{Q}$ is a partition atlas (Def. 3.2), homogeneous with contract $(C, \mathcal{T}) = (\text{topological}, \{\mathbb{R}^N : N \in \mathbb{N}\})$ (Remark 3.5).
- (3) Each fibre object Q_b has finite topological embedding dimension $\text{emb}(Q_b) < \infty$, with the type constant on each chamber so that $\text{emb}(Q_\alpha) = \text{emb}(Q_b)$ for $b \in B_\alpha$.
- (4) The base space B admits a topological embedding $\iota_B : B \hookrightarrow \mathbb{R}^{d_B}$, where $d_B = \text{emb}_{\text{top}}(B)$ is the base embedding dimension (Def. 3.12).

Assumption 4.2 (Memoryless compatibility). The compatibility structure of $\mathcal{Q}$ is *memoryless*: the overlap-compatibility conditions at levels (b), (c), (d) depend only on the points $x \in U_{\alpha\beta}$ and not on which region’s quotient is taken as reference. Equivalently, the overlap relation is symmetric, and any transition data $\phi_{\alpha\beta}$ at level (c) or above is invertible with $\phi_{\beta\alpha} = \phi_{\alpha\beta}^{-1}$.

The double pendulum satisfies both assumptions directly. Color Fantasy satisfies them only under the memoryless approximation used in Example 4.12 (thermal-mass delay treated as regime-local regression input rather than as overlap compatibility data); the full Color Fantasy atlas with hysteretic memory across regime boundaries is outside the scope of Theorem 4.8 as stated and is named as an open problem (§7.2.1). Assumption 4.1 excludes atlases with non-integer-dimension local quotients (fractal targets, quotients by non-proper actions) and bases that do not embed topologically in Euclidean

space of their covering dimension — consistent with Paper III §12.4 (Open Problem 4d), and listed in §7 as open. Assumption 4.2 excludes hysteretic and direction-dependent overlap structures — constitutive memory in the sense of [Visintin, 1994], [Mielke et al., 2002], as well as first-order phase transitions with path-dependent metastable overlaps — listed separately in §7 as open.

The role of memorylessness in the theorem. The bare proof of the inequality in Theorem 4.8 (§4.3 below) uses only the base-split coordinate factorisation (Def. 4.4) and the per-fibre faithfulness hypothesis (Def. 4.3); memorylessness does not enter Step 1 or Step 2 step-by-step. Memorylessness is required *upstream*, as the precondition under which the atlas data fits the present chamber-meeting taxonomy at all.

The taxonomy of Def. 3.1 — partition (Def. 3.2), open overlap (Def. 3.3), stratified-meet (Def. 3.4) — assumes that regime assignment is a function of state, single-valued or multi-valued depending on case but in all cases determined by $x \in \mathcal{X}$ alone. A hysteretic system breaks this: at the same point of the base, different histories can place the system in different regimes, so the regime-assignment data is neither a partition function ι nor a state-determined multifunction Λ but a path-dependent multifunction. This is a *structurally different kind of overlap* from any of (a)–(d) in §3.3: metastable coexistence regions traversed asymmetrically, ferromagnetic memory loops, rate-independent constitutive hysteresis. The compatibility-strength axis of §3.3 was not formulated to handle path-dependent overlap and the present floor theorem cannot be applied to it directly.

Bringing such a system into the present formalism requires choosing one of three remediation strategies, each producing a *different* overlap structure and *different* embedding requirements: (i) enlarge the state with internal memory variables — the augmented system is memoryless on the augmented state, but the state space is larger and per-fibre embedding dimensions rise; (ii) discretise memory into a finite set of memory states — the chamber count multiplies and the partition acquires additional disconnected components per memory configuration; (iii) work over a category of histories — the atlas is over composable trajectories rather than states, and the gluing data lives on path-overlap rather than state-overlap. The three give three different memoryless atlases with three different floors, and the choice among them is a modelling commitment whose consequences are not internal to Theorem 4.8. Memorylessness is therefore not proof scaffolding but the precondition under which the question “what is the floor?” has a single well-posed answer at all (§7.2.1).

Let $E : \mathcal{X} \rightarrow Z$ be a continuous encoder into a latent space $Z \cong \mathbb{R}^{D_{\text{ambient}}}$.

Definition 4.3 (Atlas-faithful encoder). The encoder $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ is *atlas-faithful* with respect to $\mathcal{Q}$ if:

- (1) (*Per-fibre faithfulness.*) For each $\alpha \in A$ and each $b \in B_\alpha$, the restriction of E to the fibre $\pi_B^{-1}(b)$ factors through the fibre quotient as $E|_{\pi_B^{-1}(b)} = \tilde{E}_b \circ \pi_b$, where $\pi_b : \pi_B^{-1}(b) \rightarrow Q_b$ is the fibre quotient map and $\tilde{E}_b : Q_b \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ is a topological embedding.
- (2) (*Region recognition.*) There exists a function $\rho : E(\mathcal{X} \setminus N) \rightarrow A$ with $\rho \circ E = \iota$ on $\mathcal{X} \setminus N$. The function ρ is defined on the latent shape $E(\mathcal{X} \setminus N)$, not on all of $\mathbb{R}^{D_{\text{ambient}}}$; it is continuous on each connected component of $E(R_\alpha)$ and may be discontinuous across the boundaries between $E(R_\alpha)$ and $E(R_\beta)$ in the latent shape.

- (3) (*Compatibility coherence*.) The per-fibre images $\tilde{E}_b(Q_b) \subseteq \mathbb{R}^{D_{\text{ambient}}}$ are consistent with the compatibility data of $\mathcal{Q}$ at its stated level (§3.3).

We write $\tilde{E}_\alpha : Q_\alpha \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ for the chamber-common fibre embedding when the fibre type is constant on chamber α and the embedding can be chosen independent of $b \in B_\alpha$.

Atlas-faithfulness as defined here presumes that fibre embeddings coexist within a single ambient latent space $\mathbb{R}^{D_{\text{ambient}}}$ in a way that preserves region recognition — the encoder’s image carries enough structure that a function ρ recovering chamber labels from the latent shape exists. Weaker notions admitting coordinate-entangled fibre images, in which base and fibre information are mutually inseparable inside the latent ambient, are discussed as a different representation contract in §7.3.1 and are not covered by this definition.

Definition 4.4 (Base-split atlas-faithful encoder). Let $\mathcal{Q}$ be a partition atlas (Def. 3.2) on $\mathcal{X}$ satisfying Assumption 4.1, and let $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ be an atlas-faithful encoder (Definition 4.3).

The encoder E is *base-split* if there exists a global coordinate decomposition

$$\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$$

into two linear factors, together with a topological embedding $\iota_B : B \hookrightarrow L_B$ of the base into the first factor, such that

$$\pi_{L_B} \circ E = \iota_B \circ \pi_B \quad \text{on } \mathcal{X} \setminus N,$$

where $\pi_{L_B} : \mathbb{R}^{D_{\text{ambient}}} \rightarrow L_B$ is the linear projection onto the first factor and $\pi_B : \mathcal{X} \setminus N \rightarrow B$ is the base map. The dimensions of L_B and L_F are not prescribed by the definition; they are constrained from below by Theorem 4.8.

The condition is a *global coordinate factorisation*, not merely functional base-recoverability. An encoder of the form

$$E(x) = \Psi(\pi_B(x), q_b(x))$$

in which Ψ is a continuous bijection but does not factor through linear projection onto a linear subspace of $\mathbb{R}^{D_{\text{ambient}}}$ — so the base coordinate is recoverable from $E(x)$ jointly with fibre content but not by coordinate projection alone — is *not* base-split. Such nonlinear-recoverable encoders correspond to a different representation contract (§7.3.1) and are excluded from the scope of Theorem 4.8.

Remark 4.5 (On the name ‘base-split’). The condition imposed in Def. 4.4 is a *coordinate splitting*: a global product structure $\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$ in which the first factor recovers the base coordinate independently of fibre content. Earlier drafts called this ‘separated’ by analogy with scheme-theoretic separatedness [Vakil, 2024, Ch. 11]; that analogy is suggestive but technically misleading. Scheme-theoretic separatedness is a diagonal-closedness condition and does not, in general, imply a global product structure on the total space. The condition here is strictly stronger than scheme-theoretic separatedness in the representation-learning setting: it requires the latent ambient to factor as a product whose first factor is the base coordinate. We use *base-split* for accuracy. The line-with-two-origins is still a useful intuition pump for non-base-split encoders — the encoder’s region-recognition data is locally consistent

but does not globalise to a coordinate splitting — but the categorical equivalence is not claimed.

Remark 4.6 (Scope). Definition 4.4 is stated for partition atlases at the homogeneous-topological-Euclidean contract. The overlap-atlas analogue would require the splitting to be compatible with the nerve structure rather than with a single-valued region assignment; this is part of the overlap-atlas floor open problem (§7.2.2). The stratified-meet analogue would require the splitting to be compatible with frontier data on lower-dimensional chambers; this is part of the stratified-meet open problem (§7.2.3). Theorem 4.8 below proves the partition case only.

Definition 4.7 (Atlas embedding dimensions). We distinguish two atlas-level embedding dimensions:

$$\begin{aligned} \text{emb}_{\text{top}}(\mathcal{Q}) &:= \min\{D_{\text{ambient}} : \text{there exists an atlas-faithful} \\ &\quad E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}\}, \\ \text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q}) &:= \min\{D_{\text{ambient}} : \text{there exists a base-split atlas-faithful} \\ &\quad E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}\}, \end{aligned}$$

with the convention that either is ∞ if no such E exists for any finite D_{ambient} . By construction $\text{emb}_{\text{top}}(\mathcal{Q}) \leq \text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$, since base-split encoders are atlas-faithful. Theorem 4.8 bounds $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$ from below; the gap between $\text{emb}_{\text{top}}(\mathcal{Q})$ and $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$ is part of the open problem of §7.3.1.

4.2 The base-split atlas floor theorem

Theorem 4.8 (Partition base-split atlas dimension floor, homogeneous topological contract). *Under Assumption 4.1, if a base-split atlas-faithful encoder $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ (Definition 4.4) exists for a partition quotient atlas $\mathcal{Q}$ on $\mathcal{X}$ (Def. 3.2), then*

$$D_{\text{ambient}} \geq d_B + \sup_{b \in B} \text{emb}(Q_b),$$

where d_B is the base embedding dimension (Def. 3.12) and $\text{emb}(Q_b)$ is the topological embedding dimension of the fibre quotient at b . For partition atlases with chamber-wise constant fibre type (which holds throughout this paper by Def. 3.1(4)), this reduces to

$$D_{\text{ambient}} \geq d_B + \max_{\alpha \in A} \text{emb}(Q_\alpha).$$

4.3 Proof

The proof has two steps.

Step 1 (base floor). By Def. 4.4, the linear projection $\pi_{L_B} \circ E = \iota_B \circ \pi_B$ realises a topological embedding of B into L_B via ι_B . Hence $\dim L_B \geq \text{emb}_{\text{top}}(B) = d_B$.

Step 2 (per-fibre floor). Fix $b \in B$, and consider the fibre $\pi_B^{-1}(b) \subseteq \mathcal{X}$. By the base-split condition, $\pi_{L_B}(E(\pi_B^{-1}(b))) = \{\iota_B(b)\}$, so $E(\pi_B^{-1}(b)) \subseteq \pi_{L_B}^{-1}(\iota_B(b))$. Because the splitting $\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$ is a linear direct sum, the preimage $\pi_{L_B}^{-1}(\iota_B(b))$ is the affine subspace $\iota_B(b) + L_F$, which is homeomorphic (in fact affinely isomorphic) to L_F with $\dim L_F = D_{\text{ambient}} - \dim L_B$.

By Def. 4.3(1) applied at the base point b , the restriction $E|_{\pi_B^{-1}(b)}$ factors through π_b as $E|_{\pi_B^{-1}(b)} = \tilde{E}_b \circ \pi_b$ with $\tilde{E}_b : Q_b \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ a topological embedding whose image lies in $\pi_{L_B}^{-1}(\iota_B(b))$. Hence $\text{emb}(Q_b) \leq \dim L_F$.

Combining. $D_{\text{ambient}} = \dim L_B + \dim L_F \geq d_B + \sup_{b \in B} \text{emb}(Q_b)$, taking the supremum over b in the per-fibre bound. When the fibre type is constant on each chamber, $\sup_b \text{emb}(Q_b) = \max_\alpha \text{emb}(Q_\alpha)$ and the chamber-form bound follows. $\square$

Remark 4.9 (Tightness, excess dimension, and saturation). Theorem 4.8 is a lower bound on the *base-split* atlas embedding dimension $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$ (Def. 4.7). It does not by itself prove that the floor is achievable: a matching upper-bound construction is required to demonstrate saturation. For partition atlases with regular Morse-type transitions between chambers (§3.4), the classical handle-attachment local models can be assembled within the $(D_{\text{ambient}} - d_B)$ -dimensional fibre factor, and an explicit base-split construction realising $D_{\text{ambient}} = d_B + \max_\alpha \text{emb}(Q_\alpha)$ is plausible but is not proved here in full generality; the double-pendulum atlas (Ex. 4.11) is consistent with the bound at $D_{\text{ambient}} = 5$, matching Paper VII’s empirical floor, but a rigorous upper-bound construction is open (§7.3.2).

Three distinct mathematical phenomena can affect the relationship between $\text{emb}_{\text{top}}(\mathcal{Q})$ and $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$, or move the actual encoder dimension above the base-split bound; each is a different failure mode and they are named separately in §7 rather than bundled into a single additive cost:

- (i) *Non-base-split encoders.* When the base-split hypothesis fails — the encoder admits no global coordinate splitting that recovers the base independently of fibre content — the projection argument of Step 2 does not apply. Without the splitting, the additive lower bound is not proved; the optimum $\text{emb}_{\text{top}}(\mathcal{Q})$ may be higher, equal to, or lower than the base-split bound, depending on global gluing constraints. Entangled base–fibre representations can in principle save ambient dimensions for atlases with sufficient global coupling, just as T^2 embeds in $\mathbb{R}^4$ via either a separated $S^1 \times S^1$ representation or various entangled embeddings, and global constraints determine which is achievable. Whether non-base-split encoders ever achieve genuinely lower D_{ambient} for atlas-faithful representations is open.
- (ii) *Singular transitions.* When transitions between chambers are not regular handle attachments — coinciding critical values, Morse–Bott degenerate critical submanifolds, high-index attachments that cannot be accommodated within the $(D_{\text{ambient}} - d_B)$ -fibre budget, or stratified-space singularities — the base-split bound may still hold but the construction realising it requires additional dimension beyond the clean two-term sum, placed in the fibre factor.
- (iii) *Open-overlap and stratified-meet atlases.* Theorem 4.8 is scoped to partition atlases (Def. 3.2). For overlap atlases (Def. 3.3), the gluing data on chamber intersections introduces global constraints not captured by the fibre-indexed bound; the tripod (three intervals glued at a point) is the smallest example where this matters (§7.2.2). For stratified-meet atlases (Def. 3.4), the corresponding obstruction is the absence of a base-split definition that interacts coherently with frontier data on lower-dimensional chambers; the figure-eight merge is the canonical example (§7.2.3).

Earlier drafts attempted to capture all three phenomena through a single additive term $c(\mathcal{Q}) \geq 0$. That formulation is dropped from the theorem statement because the three classes have distinct mathematical origins — failure of a hypothesis, local transition geometry, and global gluing topology respectively — and no closed-form additive cost captures them uniformly.

4.4 Corollaries

Corollary 4.10 (Single-global-quotient reduction). *When $|A| = 1$ and the base B is a single point (so $d_B = 0$), the base-split hypothesis is trivially satisfied (with L_B a point), and Theorem 4.8 reduces to $D_{\text{ambient}} \geq \text{emb}(Q)$, which is Proposition 3.1(i) of Paper III. In this case the unique fibre type $Q_\alpha = Q$ coincides with the lone quotient inherited from Papers I–III, and the legacy single-quotient setting is recovered.*

Example 4.11 (Double pendulum floor). For the atlas of Paper VII §7, the base is a bounded interval of energies, $B \subseteq \mathbb{R}$ with $d_B = 1$, and the four chambers (open intervals between Morse-critical energies) carry fibre types S^3 , $S^2 \times S^1$, $\#_2(S^2 \times S^1)$, and T^3 respectively. The topology sequence $\emptyset \rightarrow S^3 \rightarrow S^2 \times S^1 \rightarrow \#_2(S^2 \times S^1) \rightarrow T^3$ across the four critical energies follows from stratified Morse theory on the Hamiltonian sublevel sets [Milnor, 1963a, Smale, 1970a] for the equal-mass equal-link conservative model; it depends on those specific modelling choices.

For each chamber, the embedding-dimension claim $\text{emb}(Q_\alpha) = 4$ has two parts: (i) the lower bound $\text{emb} \geq 4$, which holds because no closed orientable 3-manifold embeds in $\mathbb{R}^3$ (by invariance of domain; see Appendix B.4, Lemma at the head of the section); and (ii) the upper bound $\text{emb} \leq 4$ by an explicit construction in $\mathbb{R}^4$. The per-case upper-bound constructions are given in Appendix B.4: S^3 as the unit sphere in $\mathbb{R}^4$; $S^2 \times S^1$ as the boundary of a tubular neighbourhood of an unknotted $S^2 \subset \mathbb{R}^4$; $\#_2(S^2 \times S^1)$ as the boundary of a genus-2 four-dimensional handlebody embedded in $\mathbb{R}^4$; and $T^3 = T^2 \times S^1$ as the boundary of $T^2 \times D^2 \hookrightarrow \mathbb{R}^3 \times \mathbb{R} = \mathbb{R}^4$ via tubular neighbourhood of the standard torus and the additional disc factor. Hirsch’s general theorem $\text{emb} \leq 5$ for closed orientable 3-manifolds is the fallback; the four specific cases here do better.

Theorem 4.8 then gives

$$D_{\text{ambient}} \geq 4 + 1 = 5.$$

The empirical experiments reported in Farkas [2026g] are consistent with a five-dimensional base-split representation; this does not by itself constitute a proof of saturation, since “empirical floor” is architecture- and training-dependent. A rigorous saturation construction — exhibiting an explicit base-split atlas-faithful encoder achieving $D_{\text{ambient}} = 5$ — is an open problem (§7.3.2).

Example 4.12 (Color Fantasy floor, memoryless approximation). For the Color Fantasy atlas of Paper VII §8 under its memoryless approximation (thermal-mass delay treated as regime-local regression input rather than as overlap compatibility data), the base is the joint cabin–bay temperature space $B \subseteq \mathbb{R}^2$, so $d_B = 2$. Each regime’s fibre object Q_α contains at minimum the shared day–night cyclic factor S^1 , contributing $\text{emb}(S^1) = 2$ to $\text{emb}(Q_\alpha)$; regime-specific content (delay variable, operative control subset, regression structure) contributes further dimensions. Theorem 4.8 gives

$$D_{\text{ambient}} \geq \max_{\alpha} \text{emb}(Q_{\alpha}) + 2,$$

where the $+2$ is the base embedding dimension. The precise per-regime $\text{emb}(Q_\alpha)$ values depend on the final form of Paper VII’s regime-block specifications, which is the subject of ongoing revision; the bound above is the structural form Theorem 4.8 imposes on any such specification.

Remark 4.13 (Memory scope and earlier-draft correction). *Memoryless scope.* The memoryless approximation in Example 4.12 is a scope restriction of Theorem 4.8, not of the atlas formalism. In Paper VII’s GCD architecture, the delay variable is a regime-local regression input — it lives inside Q_α rather than on the overlap between regimes — so the memoryless approximation captures exactly the scope in which Paper VII’s empirical work operates. A full atlas-floor theorem covering hysteretic memory across regime boundaries is an open problem (§7.2.1).

Year-phase variable. The available Color Fantasy data covers a single year, so the annual phase $\varphi_{\text{year}} \in S^1$ is not distinguishable in the data and does not appear in the atlas. A multi-year extension would add it as a third base coordinate, raising d_B from 2 to 3.

Earlier-draft correction. An earlier draft of this paper treated Color Fantasy as a discrete-base atlas with $A = \{R_1, R_2, R_3, R_4\}$ finite and $d_B = 0$, giving a floor of $D_{\text{ambient}} \geq \max_\alpha \text{emb}(Q_\alpha)$. That reading conflated qualitative regime labels with the underlying continuous physical variables driving the regime switches. The correct reading, used here, is that the four regimes are qualitative open subregions of a continuous two-dimensional base, structurally parallel to the double-pendulum case (Example 4.11).

Remark 4.14 (Stronger contracts, research direction). The bi-Lipschitz, smooth, and isometric analogues of Theorem 4.8 would replace $\text{emb}(Q_\alpha)$ with the corresponding per-regime floors from Paper III §3.4 and would require the base-split condition to be adapted to each contract (bi-Lipschitz splitting with bounded distortion; smooth splitting; isometric splitting). The expected form is

$$D_{\text{ambient}} \gtrsim d_B(C) + \max_\alpha \text{emb}_C(Q_\alpha),$$

under an appropriate C -base-split hypothesis, but no such theorem is proved here. The literature on stronger-contract embeddings of singular and glued spaces [Naor and Neiman, 2012, Eriksson-Bique, 2018] supports the existence of such floors when per-regime local quotients have appropriate geometric regularity (bounded curvature, doubling, bounded distortion). A full development requires assumptions stronger than Assumption 4.1 — uniform geometry across overlaps, quantitative control of transition neighbourhoods — and is deferred to §7.1.2. Classical obstructions (flat torus not C^2 -isometric in $\mathbb{R}^3$; doubling alone insufficient for bi-Lipschitz embedding into any $\mathbb{R}^N$ without further regularity, per [Lang and Plaut, 2001]) mean stronger-contract atlas floors will not generally equal the topological floor.

4.5 Remarks on what the theorem does and does not say

Remark 4.15 (What it says). The base-split atlas floor decomposes as (per-fibre topological embedding cost) + (base embedding cost): a clean sum of two terms with clear interpretations. The fibre cost is what each local quotient demands locally; the base cost is what the encoder must spend to recover the base coordinate independently of fibre content. The double pendulum and Color Fantasy illustrate the two

terms respectively — DP at $d_B = 1$ with $\max_{\alpha} \text{emb}(Q_{\alpha}) = 4$, CF at $d_B = 2$ with regime-specific fibres including a shared S^1 (Examples 4.11 and 4.12). Whether these empirical observations *saturate* the lower bound, in the sense of an explicit base-split construction realising the bound, is open (§7.3.2).

Remark 4.16 (What it does not say). Theorem 4.8 is a *lower bound* on the latent dimension required for atlas-faithful encoding. It does not say any particular encoder achieves this bound, nor that any particular training procedure produces an atlas-faithful encoder at this dimension. Those are upper-bound and constructive questions, analogous to Paper III Part III, and are not addressed here. It also does not address whether the atlas itself is correctly specified: the given-partition tier assumes $\mathcal{Q}$ is supplied. §5 addresses selection among admissible families; §6 addresses emergent discovery.

5 The learned-partition tier

§4 treated the case where the atlas $\mathcal{Q}$ is supplied — partition, local quotients, and compatibility data all given as input. Most real problems do not come with an atlas pre-specified. This section addresses the next tier: the atlas is not given, but the problem domain restricts the space of *admissible* atlases to a tractable family, and atlas selection is performed within that family.

The theoretical content is modest — no new floor theorem, because Theorem 4.8 applies per atlas hypothesis. The section’s work is in (i) defining admissible atlas families as the atlas-level generalisation of Paper VI’s admissible quotient class $\mathcal{Q}_{\text{cl}}(\mathfrak{g}, \mathcal{X})$ [Farkas, 2026f], (ii) identifying the selection criterion, and (iii) stating what evidence can and cannot distinguish atlas hypotheses.

Adjacent machine-learning frameworks that partition the input space and fit region-specific models are prior art for the learned-partition tier in part. Mixture-of-experts architectures [Jacobs et al., 1991], [Masoudnia and Ebrahimpour, 2014] have the partition-plus-local-model structure but treat local models as merely predictive. Regime-switching state-space models [Hamilton, 1989], [Kim and Nelson, 1999] formalise time-indexed latent regime variables with regime-specific dynamics. Multi-manifold learning [Elhamifar and Vidal, 2013] has the partition-plus-local-geometry structure but typically stops at clustering or low-dimensional approximation. Statistics on stratified spaces [Feragen et al., 2020] validates that singular quotient domains are standard statistical objects. What distinguishes the learned-partition tier developed here from each of these is the commitment to quotient-theoretic semantics — the local structure is an equivalence relation on the region, not a predictive model. The contrast is concrete: in a mixture-of-experts setup, two inputs are grouped because the same expert predicts them well, an operational equivalence that depends on the loss and the trained model; in a quotient-atlas setup, two inputs are grouped because the declared local equivalence relation identifies them under the regime’s structural contract, an equivalence that exists prior to any encoder being trained. The dimension-floor contract from Theorem 4.8 then constrains selection to atlases whose latent-dimension requirement is architecturally realisable. The present framework is not a replacement for the adjacent literature but a quotient-theoretic layer that can sit on top of it when atlas structure is the target.

5.1 Admissible atlas families

Definition 5.1 (Admissible atlas family). Given a problem domain, an *admissible atlas family* $\mathfrak{A}$ is a set of quotient atlases on the problem’s input space $\mathcal{X}$, each member $\mathcal{Q} \in \mathfrak{A}$ specified by:

- (1) A family of per-regime generator specifications $\{\mathbf{g}_\alpha\}_{\alpha \in A}$, each $\mathbf{g}_\alpha$ a generator specification in the sense of Paper VI (Definition 2.1 there), such that the per-regime local quotient Q_α lies in the admissible quotient class $\mathcal{Q}_{\text{cl}}(\mathbf{g}_\alpha, R_\alpha)$. Paper VI’s regularity hypotheses H1–H3 are required to hold on each R_α individually, not necessarily globally on $\mathcal{X}$.
- (2) A regime-indicator $r \in \mathcal{R}$ drawn from a domain-specific *regime-indicator class*, such that the partition $\{R_\alpha\}$ is realised as level sets or sublevel sets of r .
- (3) A compatibility level at or below a domain-specified maximum (§3.3).
- (4) The regularity hypotheses of Theorem 4.8: the atlas is a partition atlas (Def. 3.2), homogeneous with contract (topological, $\{\mathbb{R}^N\}$) (Remark 3.5), with integer-dimension local quotients and base, memoryless compatibility, and topological-embeddability of the base.

Remark 5.2 (Two specifications, coupled rather than independent). An atlas hypothesis requires specifying both the space of local quotients (via $\{\mathbf{g}_\alpha\}$) and the space of admissible partitions (via $\mathcal{R}$). These are distinguishable as mathematical objects — generator specifications describe within-region quotient structure, indicators describe region-boundary functions — but they are coupled, not independent. The regime-indicator determines the domain of validity of each $\mathbf{g}_\alpha$, and the per-regime specifications constrain which partitions are meaningful: a partition cutting a single generator-defined quotient structure into two regions is not admissible, and a generator combination requiring overlap compatibility not expressible by any $r \in \mathcal{R}$ is excluded. Admissibility is therefore a joint condition on $(\{\mathbf{g}_\alpha\}, r)$ tuples, not a product condition. This coupling is the structural source of the evidence-equivalence classes discussed in §5.4.

Remark 5.3 (Scope of Paper VI’s generator framework). Paper VI’s generator specification assumes Lie-group symmetries with free proper action on a contractible observation space (hypotheses H1–H3). Not every regime of every atlas satisfies these. Color Fantasy regimes are discrete regression structures with no Lie-symmetry content; intermediate chaotic regimes of the double pendulum are non-integrable and admit no continuous symmetry beyond time translation. Paper VI’s generator framework is therefore used within regions where it applies, and supplemented by non-Lie per-regime specifications — parametric regression families, discrete symbolic constraints, action-angle structures valid only where KAM applies — where it does not. A unified generator framework covering Lie-type and non-Lie-type regimes within the same atlas is beyond the scope of this paper (§7).

Remark 5.4 (What the regime-indicator class is). The regime-indicator class $\mathcal{R}$ is a first-class modelling object: the space of admissible partitions, specified up front and separately from the local-quotient class. $\mathcal{R}$ can be temporal (regime as a latent time-indexed state variable, as in regime-switching models), spatial (regime as a region in a spatial domain), feature-space (regime as a classification of points by their features), or mixed. Which interpretation applies must be declared as part of specifying $\mathfrak{A}$, because the search space and evidence criteria are materially different.

Change-point detection and segmentation [Truong et al., 2020] provides the cleanest prior art for $\mathcal{R}$ as a cost-plus-search-method object when $\mathcal{R}$ is temporal; partition-family model selection [Mazhar et al., 2022] is a closer formal precedent for $\mathcal{R}$ as a combinatorial object with its own prior. The distinctive contribution of Paper IV is not to propose a new search procedure but to name $\mathcal{R}$ as a component of the atlas hypothesis independent of the per-regime structure class, which this literature does not usually make explicit.

Examples of regime-indicator classes. *Continuous smooth Morse indicators.* For atlases arising from a smooth scalar function $f : \mathcal{X} \rightarrow \mathbb{R}$ with isolated non-degenerate critical points, $\mathcal{R}$ consists of all such f within a parametric family — polynomial Hamiltonians of bounded degree, energy functions with bounded control parameters. Partition is defined by the Morse-critical-value structure of f . DP-type class.

Discrete threshold indicators. For atlases arising from thresholding one or more control variables, $\mathcal{R}$ consists of thresholded classifiers on a fixed set of control variables, indexed by threshold values. CF-type class.

Mixed indicators. Combinations of the above.

5.2 Atlas selection as a joint problem

Given an admissible family $\mathfrak{A}$, the selection problem is:

Find $\mathcal{Q}^ \in \mathfrak{A}$ that best explains the observed data, subject to (i) per-regime quotient-faithfulness with each $Q_\alpha \in \mathcal{Q}_{\text{cl}}(\mathfrak{g}_\alpha, R_\alpha)$, (ii) a partition realisable by a regime-indicator $r \in \mathcal{R}$, and (iii) an encoder architecture of latent dimension at least $\max_\alpha \text{emb}(Q_\alpha) + d_B$, the base-split lower bound given by Theorem 4.8 (with additional budget potentially required for non-base-split encoders or singular transitions per Remark 4.9).*

This is a joint problem — neither partition nor per-regime specifications can be selected independently. Changing the partition changes which $\mathfrak{g}_\alpha$ applies where; changing a $\mathfrak{g}_\alpha$ changes which partitions produce coherent per-regime structure.

Remark 5.5 (Asymmetry with Paper VI’s single-quotient GCD). In Paper VI’s single-quotient GCD, the selection problem was one-dimensional: narrow $\mathcal{Q}_{\text{cl}}(\mathfrak{g}, \mathcal{X})$ by progressively refining $\mathfrak{g}$. In the atlas setting, the partition dimension is independent of the per-regime generator dimension, and the two can trade off. A finer partition with simpler per-regime $\mathfrak{g}_\alpha$ and a coarser partition with richer per-regime $\mathfrak{g}_\alpha$ can both explain the same data. This is a genuine non-uniqueness of the atlas hypothesis, following structurally from the coupling of Remark 5.2.

Remark 5.6 (Parallel work on per-region structural discovery). AtlasD [Bhat et al., 2025] is a parallel line of work addressing per-region structural discovery in an atlas framework. It formalises per-chart equivariance as atlas equivariance, trains local predictor networks per chart, and learns a Lie-algebra basis and discrete coset representatives to which the localised predictors are equivariant. Under the hypotheses that (i) the atlas on the input domain is supplied a priori and (ii) the target structure is an input-output symmetry group of the task function, AtlasD solves a version of the per-region half of the learned-partition problem. AtlasD’s stated future directions — “develop a method that discovers the atlas in tandem to the atlas equivariance group” — correspond to the joint-selection problem of the present section and to the emergent-discovery problem of §6. We therefore treat AtlasD as

closely related independent work rather than strict prior art, and we note that Paper IV’s atlas framework is broader in two directions: it addresses the co-discovery problem explicitly (§6), and it targets atlas-faithful representation structure rather than input-output task symmetry.

5.3 Evidence criteria

Three types of evidence bear on atlas selection. None is individually sufficient; the three are complementary.

(E1) Per-regime faithfulness diagnostics. For each candidate $\mathcal{Q} \in \mathfrak{A}$, train an encoder at dimension $\text{emb}_{\text{top}}(\mathcal{Q})$ (or slightly above) and evaluate per-regime quotient-faithfulness using Papers I and III diagnostics applied regionally. A candidate atlas whose per-regime faithfulness is achievable with the predicted architecture is *consistent* with the data; one whose faithfulness cannot be achieved despite meeting the predicted architecture is *inconsistent*. Necessary but not sufficient.

(E2) Region-stratified topological invariants. For each candidate $\mathcal{Q}$, compute the predicted per-regime topological invariants (Betti numbers, homotopy type, fundamental group) from each $\mathcal{Q}_{\text{cl}}(\mathfrak{g}_\alpha, R_\alpha)$. Then compute the corresponding invariants from data restricted to each region, using per-regime persistent homology — preferably level-set persistent homology on the regime-indicator r when r is smooth (the framework of Bauer et al. [2026], §4), sublevel-set PH on appropriate filtrations otherwise. §6 develops this apparatus. Stronger than (E1): compares not just whether faithfulness is achievable but whether the correct topology is recovered region-by-region.

(E3) Transition-signature diagnostics. Between adjacent regions, the local model of §3.4 predicts specific topology-change signatures at regime boundaries — handle-attachment indices for Morse transitions, discrete jumps for threshold transitions. Persistent homology computed along paths that cross a predicted boundary should exhibit the predicted signature. Mismatch indicates a misidentified boundary or a wrong local-model assumption (Morse–Bott degeneracy, non-regular transition geometry).

Remark 5.7 (Data-grading as a precondition). All three criteria presume sufficient data in each region. An atlas hypothesis that places most of the data in one regime and very little in others will have well-estimated invariants for the well-sampled regime and poorly-estimated invariants elsewhere. The stable-equilibrium-dominated DP data of Paper VII is canonical: the S^3 regime is well-estimated, the chaotic intermediate regimes are under-sampled, and selection among atlases differing mainly in their treatment of the under-sampled regimes will be ambiguous on evidence-criteria alone. A data-grading step flagging each region’s sampling adequacy, and restricting evidence-based selection to adequately-sampled regions, is a necessary prerequisite. Mechanics in §6.

5.4 What selection can and cannot deliver

What it can deliver. Given $\mathfrak{A}$, sufficient data with known per-regime adequacy, and evidence criteria (E1)–(E3), selection ranks members of $\mathfrak{A}$ by consistency and identifies a most-consistent candidate. When the true atlas is in $\mathfrak{A}$ and data is adequate across all regions, the most-consistent candidate is the true atlas up to evidence-equivalence classes.

What it cannot deliver. Selection is only as good as the admissible family. If the true atlas is not in $\mathfrak{A}$ — per-regime generator specifications are incomplete, regime-indicator class is wrong, compatibility level is misspecified — selection returns the closest member to the truth, which may not be close. No amount of within-family evidence can detect out-of-family truth. This inherits directly from Paper VI §6 and the non-identifiability results of the disentanglement literature [Locatello et al., 2019, Khemakhem et al., 2020]: identifiable recovery of latent structure requires explicit auxiliary information or structural priors, not merely observational data.

Evidence-equivalence classes. Two atlases $\mathcal{Q}_1, \mathcal{Q}_2 \in \mathfrak{A}$ predicting the same per-regime invariants and transition signatures on the data cannot be distinguished by (E1)–(E3). Such pairs form equivalence classes; the example of Remark 5.5 generates such classes generically. Selection returns an equivalence class, not a unique atlas. Resolving within-class ambiguity requires additional input: domain knowledge, auxiliary data, or explicit prior preferences (Occam-type preferences for coarser partitions, physical-interpretability preferences for specific regime indicators). §7 includes within-class resolution as an open problem.

Remark 5.8 (Relation to the emergent-discovery tier). §5 assumes $\mathfrak{A}$ is supplied. §6 addresses the case where $\mathfrak{A}$ itself is unknown and must be constructed, either by expanding a candidate family until the data is explained or by bootstrapping one from the data without a prior family. §6 inherits §5’s selection machinery as a subroutine; the new problem there is how to generate candidates without a prior family.

5.5 Examples at the learned-partition tier

Double pendulum. The admissible family $\mathfrak{A}_{\text{DP}}$ consists of atlases whose partition is defined by a smooth Hamiltonian in a parametric class $\mathcal{R}_{\text{Ham}}$, and whose per-regime generator specifications are either the local action-angle structure (where KAM-type action-angle variables exist — the low-energy S^3 regime and the high-energy T^3 regime, approximately) or trivial (in the chaotic intermediate regimes, where Paper VI’s H1–H3 do not hold and $\mathfrak{g}_\alpha$ reduces to a qualitative non-Lie specification per Remark 5.3). If the true Hamiltonian is in $\mathcal{R}_{\text{Ham}}$, evidence (E2) comparing per-regime persistent homology against the predicted topology sequence $S^3 \rightarrow S^2 \times S^1 \rightarrow \#_2(S^2 \times S^1) \rightarrow T^3$ identifies it. If the true Hamiltonian is outside the family, selection returns the closest member whose predicted topology sequence mismatches at some boundary, detectable by (E3) but not repairable within the family.

Color Fantasy. The admissible family $\mathfrak{A}_{\text{CF}}$ consists of atlases with discrete thermal partitions (four regimes) defined by thresholds on control variables, and per-regime local quotients described by regime-specific regression-model specifications. The per-regime specifications are not Paper VI generator specifications in the strict sense — no Lie-symmetry content — but fill the analogous role per Remark 5.3: they specify admissible regression structure per region. Selection identifies threshold values and per-regime model structure jointly. Paper VII §8 performed this manually using domain knowledge. A learned-tier approach would treat threshold values and per-regime model structure as simultaneously-selected; the result, if the true atlas is in the family, recovers the manual proposal.

5.6 Computational tractability

For small $|\mathfrak{A}|$, selection is direct enumeration. For large or parametric $\mathfrak{A}$, selection becomes a joint search over regime-indicators and per-regime specifications. We do not develop selection algorithms here. We observe that (i) the joint search is structurally similar to parametric model selection in statistics, which provides inherited technique; (ii) per-regime persistent homology is computable at reasonable cost with existing tools, with specifics in §6; (iii) the number of qualitatively-different atlas hypotheses in a well-chosen $\mathfrak{A}$ is typically small (DP has one parametric Hamiltonian family; CF has finitely many topologically-distinct threshold configurations). Tractability is not a primary concern for the theoretical framework and becomes one when applied at scale.

6 The emergent-discovery tier

§5 assumed an admissible atlas family $\mathfrak{A}$ was supplied and selection was performed within it. This section addresses the harder case where $\mathfrak{A}$ itself is unknown and must be constructed from data — or, in the strongest version, where no prior atlas hypothesis is available and both the partition and the per-region local structure must be co-discovered.

This section develops the framework for emergent discovery. It does not claim the problem is solved. The honest claim is narrower: given the apparatus of Papers I–VI plus §§3–5, a specific three-component structural gap separates the present state of the art from a rigorous emergent-discovery procedure. The section names the three components, discusses what is known about each, and presents a *portfolio* of diagnostic tool classes available for partial progress — not a unified pipeline. Different tools attack different slices of the problem, with different assumption strengths, different readiness levels, and different natural contract levels. Writing this section as a menu rather than as a single procedure is the honest reflection of the state of the field.

6.1 The co-discovery problem

Emergent discovery faces a chicken-and-egg structure absent from the given and learned tiers.

At the given tier (§4), the partition $\{R_\alpha\}$ and local quotients $\{Q_\alpha\}$ are both supplied, and the task reduces to per-region quotient-faithful encoding plus compatibility-coherent region recognition.

At the learned tier (§5), the partition is drawn from a regime-indicator class $\mathcal{R}$ and the local quotients from admissible classes $\mathcal{Q}_{\text{cl}}(\mathfrak{g}_\alpha, R_\alpha)$. Both are variable, but both are drawn from supplied classes, and selection proceeds jointly.

At the emergent tier, no $\mathcal{R}$ or $\{\mathfrak{g}_\alpha\}$ is supplied. The data alone must yield candidate partitions, and within each candidate partition, candidate local quotients. Both have to be estimated from the same data, simultaneously, without either being pre-specified.

This is genuinely harder than selection with priors because the coupling from Remark 5.2 now works against discovery: you cannot estimate the partition without knowing what structure sits in each region, and you cannot estimate per-region structure without knowing the partition. Within-regime persistent homology requires specifying the regime boundary; correct regime identification requires knowing what topological signature to look for.

We note honestly at the outset that topological data analysis (TDA) is currently the most crystallised diagnostic tool available for atlas-structure discovery. This is a contingent

fact about the development of the field, not a statement that TDA is the appropriate framework for emergent atlas discovery in general. Other tool classes — dynamical reconstruction, spectral analysis of learned representations, mechanistic interpretability, symmetry discovery, causal-structure discovery — are each at earlier stages of development and offer complementary diagnostic power. §6.3 presents them in parallel as a portfolio rather than through the TDA lens alone. The framework of this section is deliberately agnostic about which tool class is used; it describes the structural gap and the kinds of evidence any tool needs to provide to close it.

Paper IV does not solve the co-discovery problem. What it does is name three specific structural components without which a rigorous solution is not available, identify the diagnostic tool classes that partial solutions can build on, and state readiness assessments honestly for each.

6.2 The structural gap: three missing components

Closing the co-discovery problem rigorously requires three components, each of which is currently unresolved in the representation-learning literature. The structure of the three is clarified by an analogue from pure mathematics: the sheaf-theoretic construction of shape space [Arya et al., 2025], which constructs a faithful summary object for shapes via a site-theoretic equivalence. A shape is recovered homeomorphically from its faithful-summary site under a subcanonicity condition. The analogy is methodological — it identifies what kind of object would close the gap and what conditions it would need to satisfy — not an identification of neural representations with shape-space sheaves. For a neural analogue of the emergent-discovery tier, the three components are:

(A) A neural-side faithful-summary object. A diagnostic for atlas structure in a learned representation requires a summary object that (i) is *faithful* — captures enough structure to distinguish correct from incorrect atlas hypotheses up to the relevant notion of equivalence — and (ii) is *computable* from the network’s output on data, not from privileged knowledge of the data-generating process. Arya et al. [2025] demonstrate such an object for shapes: a sheaf-glued persistent-homology transform (PHT) that is injective up to shape isomorphism and assembles from local data via a descent construction. For a neural analogue, three specific properties are required:

- *Representability as a site*: the summary object must have enough internal structure to support sheaf or descent-style assembly.
- *Realisability or subcanonicity*: the assembled summary must recover the represented object up to the desired equivalence, rather than capturing only invariants.
- *Training-dynamics compatibility*: the summary must behave well under the perturbations induced by gradient training — persistent across seeds and architectures within a class, while remaining sensitive to genuine structural differences.

Candidate building blocks for such an object exist — local Jacobian spectrum, activation-region adjacency data, centroid-type local summaries, persistent features across training perturbations, learned covers — but no current candidate has been shown to satisfy all three properties jointly.

(B) A neural-side base poset or site. The sheaf-theoretic descent used in the shape case requires a base poset or site over which the local summaries are indexed. For shapes this is the ambient space. For neural representations the corresponding object is unclear. Candidates: the activation-region arrangement and its face lattice for

ReLU networks [Montúfar et al., 2014, Balestrierio and Baraniuk, 2018]; neuron-subset lattices; data-neighbourhood posets; sparse-routing decompositions. Each has structural properties relevant to different aspects of atlas discovery. A principled choice has not been made. The bottom-up theory of native neural representational grammar being developed in Paper 8 (Farkas, *Neural Realizability of Latent Structure*, forthcoming) will supply candidate objects from first principles. Paper IV therefore identifies (B) as requiring bottom-up neural-representation theory and forward-references Paper 8 rather than speculating here.

(C) Training-dynamics compatibility. The sheaf-theoretic assembly mechanism assumes that local summaries agree on overlaps in a well-defined sense. For neural representations under gradient-based training, “local” varies over training, overlaps shift as activation regions migrate [Humayun et al., 2023], and the notion of “agreement on overlap” requires a training-respecting compatibility condition. What metric, what notion of closeness, what type of descent behaviour under gradient updates — these are open. Recent work on stable and steerable sparse autoencoders [Karvonen et al., 2024] and on representational-similarity diagnostics across seeds [Klabunde et al., 2023] establishes that seed-variance and architecture-variance in learned features are substantial and measurable. This is the empirical observation that any (C)-solving framework must engage with; it is not itself a solution.

The three components are not independent. A choice of (A) may force or constrain choices of (B) and (C). The current state is that no complete triple is known for emergent atlas discovery in learned representations.

Remark 6.1 (Honest position). Paper IV identifies these as the structural gap, not as solved problems. The claim is that rigorous emergent atlas discovery requires all three, and that partial diagnostic progress — developed below — can proceed without the full triple by making specific limited assumptions about each component.

6.3 A diagnostic portfolio

Tool classes available for partial progress toward emergent discovery, organised by natural contract level and readiness. Each subsection names the class, describes what it is good at, states limitations, and gives a readiness assessment. No ordering is implied — the classes are complementary and which is most useful depends on the problem.

6.3.1 Topological and sheaf-based diagnostics

The most mature diagnostic family for structural questions of the kind atlas discovery asks. Core objects: persistent homology (one-parameter and multiparameter [Lesnick, 2015]), level-set persistence, extended persistence, derived-sheaf formulations [Curry, 2015], learned covers [Scoccola et al., 2025], cover-based complexes.

What it is good at. Detecting topological signatures of candidate regimes (Betti numbers, persistent features); comparing observed topology against predicted topology under an atlas hypothesis; identifying candidate regime boundaries via discontinuities in level-set persistence along a proposed regime-indicator; providing stability guarantees under input-data perturbation for single-parameter PH [Edelsbrunner et al., 2002, Carlsson, 2009] and improving stability for density-sensitive bifiltrations and multiparameter cases [Blumberg et al.].

Specific results worth citing. Level-set persistent homology (framework of [Bauer et al., 2026](#), §4) provides the right formalism for atlases arising from a smooth regime-indicator r : the persistence diagram of $\{r^{-1}(a)\}_{a \in \mathbb{R}}$ has bottleneck-stable signatures at topological-transition energies, exactly what the DP case of §2.2 requires. Cover learning [[Scoccola et al., 2025](#)] provides learned, topologically-faithful simplicial models that are near-precedents for atlas-adaptive complexes (§7). For stratified-space inference specifically, persistent stratified homotopy types [[Mäder et al.](#)] is directly relevant.

Limitations. Topological diagnostics capture topology only. They cannot detect structural differences that preserve topology (e.g., two regimes with the same fibre topology but different dynamical content). The no-go theorem for group-valued invariants of multiparameter persistence ([Bauer et al., 2026](#), Appendix A.2) places a principled ceiling on what additive invariants of multiparameter persistence can stably measure. Honest caveat: stable topological summaries can fail to carry the inferential information one hopes for [[Chen et al., b](#)].

Readiness. Ready to deploy for candidate-partition diagnostics, cover audits, and scale-sensitive stability analysis. Partially ready for true emergent atlas discovery in the full sense of §6.2.

Natural contract level. Topological to cover/sheaf.

6.3.2 Dynamical and spectral diagnostics

For problems where the data has dynamical structure (time series, trajectory data), dynamical-systems reconstruction methods provide a different class of diagnostic.

What it is good at. Takens-type delay embeddings [[Takens, 1981](#), [Sauer et al., 1991](#)] reconstruct a state-space within a smooth deterministic regime with provable faithfulness. Homogeneous differentiators [[Sutulovic et al., 2025](#)] de-noise measurements and estimate derivatives with rigorous error bounds, enabling attractor reconstruction under noise. Koopman and DMD methods [[Kutz et al., 2016](#), [Brunton et al., 2016](#)] identify linear-evolving observables, which can surface regime structure via changes in modal spectra. Reservoir-computing approaches provide an internal diagnostic for faithful reconstruction through the conditional Lyapunov exponent [[Hart et al.](#)] — successful attractor reconstruction requires the reservoir’s maximal conditional Lyapunov exponent to be sufficiently negative relative to the target system.

Limitations. The strongest theorems live in smooth, single-regime, deterministic dynamics with observability assumptions. Hard regime switches, genuinely stratified phase spaces, and noise models that encode real structural discontinuity (not merely smooth noise) are not directly covered. Applying homogeneous-differentiator preprocessing to data with genuine regime boundaries risks smoothing over the boundaries themselves — though this risk can be turned into a diagnostic: regions where reconstruction is unstable become candidate boundaries. Attractor reconstruction and regime discovery are often conflated in practice; many methods reconstruct a faithful state-space *within* a regime without identifying where the regime boundaries are.

Readiness. Partially ready, especially for time-series-heavy problems where local laws are genuinely dynamical. The reservoir-computing internal-diagnostic mechanism is ready now for testing specific atlas hypotheses.

Natural contract level. Dynamical; some spectral diagnostics reach bi-Lipschitz or metric contract.

6.3.3 Information-theoretic, causal, and mechanistic diagnostics

For problems where the relevant structural question is factorisation, conditional independence, or causal dependence, rather than topology.

What it is good at. Information-theoretic diagnostics [Tishby and Zaslavsky, 2015] evaluate whether a representation is compressive, approximately factorised, or has useful conditional-independence structure. Causal-discovery frameworks [Zheng et al., 2018] select among admissible causal structures given observational or interventional data. Context-specific causal models — CStreets [Duarte et al.] — explicitly generalise to region-dependent factorisation and local conditional independences, which is structurally close to atlas-level regime-dependent structure. Mechanistic-interpretability tools (sparse autoencoders, activation patching, probing classifiers, circuit analysis) provide per-region hypothesis generation and causal-intervention-based falsification — patching within a candidate region tests whether the candidate chart is mechanistically real [Saphra et al.].

Specific results worth citing. The Locatello et al. [2019] unidentifiability theorem for disentanglement is the foundational impossibility statement for unsupervised latent-factor recovery without auxiliary structure. Khemakhem et al. [2020] is the canonical identifiability-under-auxiliary-variables counterpart. These results translate directly to the atlas setting: identifiable recovery of per-region structure requires supplied or implicitly-present side information (regime labels, contexts, interventions), not observational data alone. Per-region probing classifiers and activation patching are the natural mechanistic-interpretability analogues — they operate *within* a candidate partition rather than across it.

Limitations. Information-theoretic methods usually assume globally factorised structure; the atlas framing is explicitly comfortable with different factorisation patterns in different regions. Causal discovery assumes the relevant variables and contexts are already meaningfully specified. Mechanistic-interpretability features are basis-dependent, and sparse-autoencoder features are now known to vary materially across seeds and training choices [Karvonen et al., 2024]. These methods generate local hypotheses and falsify specific claims; they do not furnish the base poset or the faithful-summary object of §6.2.

Readiness. Partially ready where contexts, interventions, auxiliary variables, or regime labels already exist or can be approximated. Ready now for local hypothesis generation and causal falsification of per-region claims.

Natural contract level. Probabilistic / causal / feature.

6.3.4 Multi-manifold and symmetry-discovery diagnostics

For problems where the relevant structural question is “are there distinct local manifolds or local symmetry groups?” rather than topology.

What it is good at. Sparse subspace clustering [Elhamifar and Vidal, 2013] and union-of-manifolds methods propose candidate partitions when data looks like a union of locally simple components. Local intrinsic-dimension estimation [Ceruti et al.] is an underrated region-change diagnostic: if a candidate partition yields stable, interpretable, region-wise intrinsic dimensions, that is non-trivial evidence for the hypothesis. Deep clustering extends this logic to nonlinear structure. For symmetry discovery specifically, AtlasD [Bhat et al., 2025] is the direct precedent for per-region structural discovery in an atlas framework: given an atlas on the input domain, discover per-chart equivariance to a Lie group plus coset representatives of the discrete component. AtlasD is independent, close-parallel work to Paper IV’s learned-partition tier (Remark 5.6), and its apparatus can be used as a per-region diagnostic within an atlas hypothesis.

Limitations. Multi-manifold methods are better at proposing candidate partitions than at deciding how those partitions assemble into an atlas. Clustering methods track density modes or semantics rather than local coordinate laws; a cluster is not a chart. Symmetry-discovery methods in general focus on observed input–output maps rather than hidden representational structure, and most are restricted to global symmetry. AtlasD is the exception in targeting local symmetry, but still discovers symmetry of the task function, not of the learned representation’s internal structure, and requires the atlas on the input domain to be supplied a priori.

Readiness. Multi-manifold and intrinsic-dimension methods are ready now for candidate partition discovery and first-pass region proposals. AtlasD is ready now for per-region symmetry discovery given a supplied atlas. Joint atlas-plus-symmetry discovery remains an open problem, flagged by AtlasD itself and corresponding to Paper IV’s emergent tier.

Natural contract level. Local linear/manifold; symmetry/equivariance for AtlasD-style diagnostics.

6.3.5 Perturbation-robustness and data-grading diagnostics

Meta-diagnostics that evaluate the reliability of the structural claims produced by the preceding tool classes.

What it is good at. On the topological side, perturbation-robustness tools are mature: confidence sets for persistence diagrams [Fasy et al., 2014], bootstrap procedures for persistence diagrams and landscapes [Chazal et al.], Bayesian frameworks for persistent homology [Maroulas et al.], subsampling confidence bounds for persistence diagrams built from time-delay embeddings with finite-sample validity under positive-reach assumptions [Niyogi et al., 2008]. Data-grading via reach estimation [Aamari et al., a] and universally-consistent reach estimation [Aamari et al., b] provides a formal regularity check on candidate regions. Local persistent homology and local intrinsic dimension [Stolz et al.] offer region-wise complexity diagnostics. Conformal prediction [Angelopoulos and Bates, 2021] calibrates downstream decisions but does not directly certify atlas

claims. Sparsification tools [Sheehy, 2013, Boissonnat and Pritam, 2020, Chazal et al., 2009] make scale-level robustness assessments computationally tractable.

Specific mechanics. A confidence-scoring procedure for atlas-structure claims must name its *perturbation class* explicitly. Too narrow (same architecture, same optimiser, only seed variation) measures robustness-within-regime rather than structural truth. Too broad (cross-architecture, cross-paradigm) may make everything look unstable. The choice is a methodological commitment and must be stated. The topological-side literature has developed mature perturbation-class conventions (seed variation, bootstrap resampling); the representation-learning side has not, and conclusions drawn from single perturbation classes should be caveated accordingly.

Limitations. Topological perturbation-robustness is strong for individual PH summaries but weaker for full structural atlas claims. The representation-learning side is genuinely under-developed: seed-variance studies show training perturbations produce materially different internal solutions [Klabunde et al., 2023]; sparse-autoencoder feature decompositions vary across seeds [Karvonen et al., 2024]; no standard probability can currently be attached to a complete atlas claim. Any such probability should be treated as heuristic unless it lives at the level of a well-defined statistic or posterior model.

Readiness. Partially ready. The topological half is mature; the representation-structural half is early.

Natural contract level. Meta-diagnostic; inherits the contract level of the underlying tool.

6.3.6 Readiness summary

Consolidating the above assessments. Readiness claims reflect the state of the literature as of the drafting period; the field is moving quickly enough that tools labelled “partially ready” could mature substantially within the series’ publication window.

Tool class	Candidate generation	gen-	Hypothesis testing	test-	Contract
Topological/sheaf	ready		ready (topological content)		topological cover/sheaf to
Dynamical/spectral	partial		ready (dynamical regime interior)		dynamical/metric
Information/causal/mech.interp	partial (with contexts)	con-	ready (per-region, given partition)		probabilistic/feature
Multi-manifold/subspace	ready		partial		local linear/manifold
Symmetry discovery (incl. AtlasD)	research direction		ready (per-region, given partition)		symmetry/equivariance
Perturbation-robustness	partial		ready for PH-level claims, early for structural		meta
Sampling-adequacy/reach	ready (diagnostic)	(pre-	ready (precondition)		geometric precondition

6.4 What the portfolio enables and what remains missing

The portfolio of §6.3 enables a pragmatic but limited workflow. Given data and a domain where some diagnostic-tool class is applicable:

- Propose candidate regions using multi-manifold or intrinsic-dimension diagnostics; if a plausible regime-indicator r exists from domain knowledge, use level-set persistent homology on r to identify candidate boundaries.
- Within each candidate region, apply per-region persistent homology, per-region mechanistic-interpretability probes, or per-region symmetry discovery (AtlasD-style) to characterise local structure.
- Apply perturbation-robustness diagnostics to separate stable structural features from training artifacts.
- Data-grade each region: adequate sampling, local regularity, per-region reach-like diagnostics.
- Evaluate the candidate atlas hypothesis against Theorem 4.8’s floor (architecture capacity at $\text{emb}_{\text{top}}(\mathcal{Q})$) and against region-stratified topological invariants.

This produces an atlas hypothesis with explicit uncertainty quantification: which regions are well-supported, which are ambiguous, where the hypothesis is architecture-consistent, and where it fails at predicted boundaries.

What the portfolio does not yet produce is a rigorous *certification* of emergent atlas discovery. Persistent-mismatch between the hypothesis and the data is not yet guaranteed to be diagnostic of specific atlas-structure features — the mismatch may be a training artifact, a data limitation, or a subtle sampling effect. The missing step is closing the loop: demonstrating that observed mismatch under a faithful-summary diagnostic (§6.2 component (A)) on a neural base poset (component (B)) with training-dynamics

compatibility (component (C)) is necessary and sufficient for atlas-structure failure. Paper IV does not close this loop. It identifies the gap structurally and names what closing it would require.

We note specifically that Paper VII’s observation — single-barcode persistent homology on the full double pendulum trajectory data reflects only the homotopy type of the sampled phase-space region (retracting onto T^2), not the per-regime dynamical topology — has a concrete procedural response within the §6.3 portfolio. Level-set persistent homology on energy (as candidate regime-indicator from domain knowledge) identifies candidate boundaries at the four Morse-critical values; per-regime persistent homology within each candidate regime produces the predicted per-regime invariants. This is a usable workflow given that energy is a plausible regime-indicator, not an emergent-discovery algorithm that would derive energy as the correct indicator without domain knowledge.

6.5 Contract-level considerations at the emergent tier

A distinction from Papers III and §4.4 worth making explicit at the emergent tier. Theorem 4.8 is stated at the topological contract. Stronger contracts (smooth, bi-Lipschitz, isometric) give stronger floors. At the emergent tier, contract choice affects both the floor and the diagnostic: different contracts distinguish different structures. The topological contract preserves topology but not distance; topological diagnostics are correspondingly blind to metric or dynamical content. The bi-Lipschitz contract preserves distances up to distortion; diagnostics at this level include metric-space-level persistent invariants and spectral structure. The isometric and physics-loss contracts preserve more; diagnostics include curvature and conserved-quantity comparisons.

An observation from within the series. Paper VII’s double pendulum experiment uses a Hamiltonian loss $\mathcal{L}_{\text{phys}}$ and its Color Fantasy experiment uses thermal-physics-loss constraints. Neither paper frames this explicitly as contract-raising. But the physics losses function as exactly that: they demand that the encoder respect specific physical relationships — energy conservation, thermal-regime-specific dynamics — that are not topological content and not captured by topology-level faithfulness. Empirically, $D_{\text{ambient}} = 5$ with Hamiltonian loss achieves $2.5\times$ lower canonical-momentum prediction error on the double pendulum than $D_{\text{ambient}} = 3$ without it (Paper VII, DP results section). The gap is not a dimension-floor phenomenon (Theorem 4.8 predicts $D_{\text{ambient}} \geq 5$ for the DP atlas at the topological contract and Paper VII’s $D_{\text{ambient}} = 5$ models meet this); it is a contract-level phenomenon. The physics loss operates at a structure-faithful contract that is stricter than topological faithfulness but generally incomparable with bi-Lipschitz and isometric contracts (per Remark 3.5’s branched-hierarchy discussion), and Paper VII’s success depends on it.

This means that the implicit contract level of the empirical work in Papers II, V, VI is already outside the purely topological setting of Paper IV’s §4, and the single contract hierarchy of Paper III is itself only one branch. Atlas floors at the Archimedean stronger contracts (§7.1.2), at the structure-faithful physics contracts (§7.1.3), and at the non-Archimedean ultrametric contract (§7.1.5) are each separate open problems; the empirical record already indicates that at least the structure-faithful direction is operationally relevant. A full atlas theory must eventually include them. We do not attempt that here.

Physics-informed learning [Karniadakis et al., 2021], equivariance-aware architectures [Weiler and Cesa, 2019], and topology-aware losses [Hofer et al.] are all instances of the

same phenomenon: they function as contract-raising mechanisms that shrink the space of representations training can realise. The ML literature describes them as inductive biases, priors, or architectural constraints. Paper IV’s contract-hierarchy language is sharper; we signpost this as a reframing rather than claiming a derivation from existing literature.

7 Open problems

Nineteen open problems, grouped into five thematic clusters: dimension-theory and contract extensions; gluing, overlap, and transition structure; representation-contract and saturation problems; emergent discovery and learning dynamics; and directions beyond the quotient universe. Each entry states the problem, summarises partial progress in the literature where relevant, and identifies what would constitute a solution.

7.1 Dimension-theory and contract extensions

These problems extend the proved partition-base-split theorem along the dimension-theoretic and contract axes: relaxing integer-dimension assumptions, raising the regularity contract above topological-Euclidean, and admitting target classes outside the Archimedean linear chain.

7.1.1 Non-integer-dimension quotients and bases

Theorem 4.8 requires integer covering dimension for both local quotients Q_α and the base A , excluding fractal or self-similar quotient structures, non-proper group actions, and atlases with fractal-dimension bases. This is consistent with Paper III §12.4 (Open Problem 4d) for the single-quotient case. Measure-theoretic extensions of delay embedding [Berry et al.] and modern fractal-dimension reviews [Fraser et al.] provide partial progress for specific cases but no general atlas-floor theorem. A solution would require either a fractal-dimension-aware embedding theory for glued spaces (extending Naor–Neiman snowflake embeddings to stratified or glued settings) or a structural reformulation of Theorem 4.8 in terms of Assouad-type or Hausdorff dimension with corresponding floor statements.

7.1.2 Stronger-contract atlas floors (Archimedean family)

Remark 4.14 flags the bi-Lipschitz, smooth, and isometric atlas floors as research directions. Paper III’s contract hierarchy (§3.4) applies per-region; the extension to the full atlas setting is an open problem at each contract separately, with the literature mature enough to identify what is missing but not to supply it.

Bi-Lipschitz atlas floors. The Lang–Schlichenmaier finite-union theorem [Lang and Plaut, 2001] gives $\dim_N(Y \cup Z) = \max\{\dim_N Y, \dim_N Z\}$ for Nagata dimension, and Makarychev–Makarychev give a two-piece Euclidean union theorem with distortion bound $7D_A D_B + 2(D_A + D_B)$ and a $3 - \varepsilon$ distortion lower bound. Seo’s local-to-global theorem constructs global bi-Lipschitz embeddings from a closed embedded singular set together with uniform Whitney-cube charts on the complement. What is missing is a finite-family Euclidean bi-Lipschitz theorem whose target dimension is of the form $\max_\alpha N_\alpha + O(\text{overlap complexity})$: the available theorems either give Nagata-dimension control without Euclidean conclusion, or Euclidean conclusion with additive dimension cost that does not reflect overlap nerve structure. Le Donne–Rajala’s tangent-dimension

theory shows that uniformly close tangents do not suffice for local-to-global dimension recovery without uniformity of tangent constants; this is a precise obstacle to naive atlas-lifting. Quantitative positive results exist for bounded-curvature orbifolds [Eriksson-Bique, 2018], quasiconformal trees, and snowflake embeddings of doubling spaces [Naor and Neiman, 2012], but none packages as a sharp atlas floor theorem. Open.

Smooth atlas floors. The general realisation theorem for abstract stratified sets (Natsume; extended in modern terms by Nocera–Volpe with conically smooth atlases, and reaffirmed for s -manifolds-with-corners by Joyce) is $D_{\text{ambient}} \leq 2n + 1$ for n -dimensional abstract stratified sets, and this bound is reported as minimal in the general stratified setting. No Haefliger-style improvement of the form $2n - k$ under connectivity assumptions on strata or links has been proved for stratified targets; the modern atlas theory [Nocera and Volpe, 2023] is intrinsic and structural rather than quantitative in Euclidean target dimension. Ayala–Francis–Tanaka’s unzipping and open handlebody decompositions provide exactly the structural tools one would need for a stratified Haefliger programme but stop short of the dimension theorem itself. A sharp smooth-contract atlas floor — depending on per-region Whitney/Haefliger bounds, link dimensions, and stratum connectivity — is open.

Isometric atlas floors. The two-dimensional case is strong: every convex Euclidean cone-metric on S^2 is realised by a unique convex polyhedron in $\mathbb{R}^3$ (Alexandrov); every compact orientable PL 2-manifold admits a PL isometric embedding in $\mathbb{R}^3$ (Burago–Zalgaller); and flat tori of any modulus admit explicit origami isometric embeddings in $\mathbb{R}^3$. In higher dimension the picture changes: Saucan proves that the Burago–Zalgaller method does not extend to $n \geq 3$ PL manifolds with codimension one, and Hartman–Nirenberg / Massey flat rigidity shows that the C^2 category is strictly more constrained than the PL category already for the flat torus. The Nash–Kuiper / Nash–Günther manifold floors (Paper III §3.4) extend to stratified targets only in the sense that boundary-extension theorems [Natsume, 1980] reduce the boundary case to the boundaryless smooth case. A general isometric atlas floor for finite-dimensional Alexandrov spaces, Riemannian orbifolds, or stratified Riemannian spaces with controlled singular strata is not established. Open.

Summary. A complete stronger-contract atlas floor theorem — with per-region geometric assumptions explicit and the base-split hypothesis adapted to each contract — is open at each of the bi-Lipschitz, smooth, and isometric contracts separately. Empirically relevant per §6.5; theoretically open.

7.1.3 Physics-contract atlas floors and quantitative bridges

The structure-faithful contracts named in Remark 3.5 — symplectic, Hamiltonian-conjugacy, Lagrangian, equivariant, measure-preserving, hard-constrained operator — are not a linear refinement of the bi-Lipschitz or isometric contracts. A linear symplectic map $\text{diag}(\lambda, \lambda^{-1})$ is symplectic but has unbounded condition number, so symplecticity does not imply a uniform bi-Lipschitz bound; conversely, a bi-Lipschitz map need not preserve any symplectic, Hamiltonian, or equivariant structure. The contracts are generally incomparable.

Three concrete open directions for atlas floors at physics contracts:

1. *Symplectic / Hamiltonian atlas floors.* A closed symplectic manifold cannot symplectically embed into exact Euclidean space of any dimension (Tischler), so the

natural target class is $\mathbb{CP}^N$ with the Fubini–Study symplectic form, and the floor becomes cohomological rather than purely dimensional. The stabilised existence result (Tischler–Gromov) identifies $\text{SympEmb}(M, \mathbb{CP}^\infty)$ with the integral-cohomology-selected component of the smooth embedding space up to weak homotopy equivalence. An atlas-level analogue — for a partition atlas in which each Q_α carries a symplectic structure and the target is a single $\mathbb{CP}^N$ — is not established.

2. *Equivariant atlas floors.* Wang’s finite-group theorem gives a $d|G| + 1$ upper bound for equivariant embedding dimension and a 2ℓ lower bound for cyclic actions with pairwise-coprime orbit lengths; Gutman et al.’s equivariant Menger–Nöbeling theorem gives a $\dim X_{\leq N} < rN/2$ condition for shift-type targets, with matching sharpness. These are single-region results. The atlas analogue, with different orbit-type strata across regions, is not established.
3. *Quantitative bridges between physics and metric contracts.* Sackel–Song–Varolgunes–Zhu’s quantitative non-squeezing result bounds the volume defect $\text{Vol}_4(E(\Phi)) \geq c(R)/L^2$ for a symplectic embedding with Lipschitz constant L , with a matching $C(R)/L$ upper bound via explicit construction. Together with Viterbo’s capacity-volume inequality, this provides a *hybrid contract* — symplectic plus Lipschitz cap — that sits between the physics and metric families rather than forming a linear chain. An atlas-level hybrid-contract floor theorem, with quantitative non-squeezing bridges calibrating symplectic rigidity against bi-Lipschitz distortion budgets, is an open problem.

7.1.4 Heterogeneous-contract atlas floors

Definition 3.1 admits atlases in which distinct regions carry distinct contracts — a region R_α with smooth dynamical content ($C_\alpha = \text{smooth}, \mathcal{T}_\alpha = \text{Euclidean}$) adjacent to a region R_β with discrete hierarchical structure ($C_\beta = \text{ultrametric}, \mathcal{T}_\beta = p\text{-adic}$), or a region with symplectic structure adjacent to a region with bi-Lipschitz structure. The mathematical theory of such heterogeneous-contract atlases is not developed here and represents a genuine gap in the existing literature.

The scaffolding exists. Vistoli’s fibred-category formalism [Vistoli, 2005] and the Stacks Project’s stack conditions handle regionwise-varying local categories with effective descent. Treumann’s constructible stacks on a stratified space, and the striation-sheaf theory for ∞ -categories, handle stratified variation with exit-path descent. Shende–Takeda’s constructible sheaves and cosheaves of dg categories realise categorical gluing with recollement and Calabi–Yau invariants. Perverse schobers model interface data as spherical functors. The variable-exponent, variable-smoothness, and wedge-Sobolev analytic traditions handle spatially varying regularity as a single global object. None of these frameworks, as far as I can determine, packages the cross-species comparison calculus on overlaps that a heterogeneous-contract atlas would require: they handle varying *fibres* of one species, not *varying species* glued along comparison morphisms.

The load-bearing new mathematics would be a theory of comparison functors, bimodules, profunctors, or mediator categories on overlaps between regions carrying unlike contracts, together with a descent condition that produces global objects and obstruction classes. Concrete test cases: (a) a base with a smooth region and a bi-Lipschitz region meeting along a lower-dimensional stratum, (b) a mixed Euclidean/ p -adic atlas of the kind Martins hints at under the "adelic predictors" heading but does not formalise, (c) a defect-TQFT-style atlas where strata of different codimension carry distinct phase-label species. An embedding-dimension floor theorem for heterogeneous atlases is open

at the level of foundations.

7.1.5 Ultrametric and p-adic atlas floors (non-Archimedean family)

The non-Archimedean target family named in Remark 3.5 is a parallel branch to the Archimedean contracts, not a refinement. A metric space is bi-Lipschitz embeddable into an ultrametric space iff it is uniformly disconnected, while compact ultrametrics of infinite Assouad dimension admit no bi-Lipschitz embedding into any finite-dimensional Euclidean space. So the Archimedean and ultrametric contracts are rigorously incompatible: neither implies the other, and they govern disjoint classes of faithful encodings.

Recent foundations exist but are uneven. Kiss–Pál prove a sharp minimal-width universal approximation theorem for p-adic ReLU networks on $\mathbb{Q}_p^{d_x} \rightarrow \mathbb{Q}_p^{d_y}$. N’guessan et al. give the *Finite Hierarchical Approximation Theorem* for van der Put networks: a depth- K v-PuNN with $\sum_{j=0}^{K-1} p^j$ neurons represents any K -level tree exactly [N’guessan et al., 2025, Martins, 2026]. Ishiki’s non-Archimedean Arens–Eells theorem embeds every ultrametric space isometrically into some (generally infinite-dimensional) valued-field extension. Bradley’s dendrogram construction realises hierarchies in the Bruhat–Tits tree of a suitable p-adic field.

Two open problems are genuinely two levels deep:

Single-region ultrametric floor. No finite-dimensional $\mathbb{Q}_p^d$ embedding-floor theorem exists for ultrametric quotients — no theorem analogous to Paper III’s Proposition 3.1 giving the minimum d for isometric or bi-Lipschitz embedding of a given ultrametric space into $\mathbb{Q}_p^d$. Existence via Hahn-field extensions is known but infinite-dimensional. A candidate conjectural form — $d \gtrsim \log_p |X|$ for a finite ultrametric X , since $\mathbb{Q}_p^d$ has only p^d balls of a given radius — is not proved.

Atlas-level ultrametric floor. Given an unresolved single-region theorem, the atlas analogue is necessarily open too: a partition atlas whose local quotients Q_α are ultrametric spaces, mapping into $\mathbb{Q}_p^m$ or into a product of p-adic target spaces, with a base A that is itself either ultrametric or mixed. The structure-theoretic question is whether the Archimedean two-term decomposition (per-region cost + base cost) has a non-Archimedean analogue, or whether the ultrametric target class forces qualitatively different floor terms — possibly involving tree branching factors, residue-field sizes, or ball-poset complexities.

Both problems are open. The current literature supports naming the ultrametric contract as a target class in Def. 3.1 but not proving any ultrametric analogue of Theorem 4.8.

7.2 Gluing, overlap, and transition structure

These problems address chamber-meeting cases the partition theorem does not cover — hysteretic memory, open-overlap nerve structure, and stratified-meet frontier compatibility — each requiring its own definitional machinery before a floor theorem can be stated.

7.2.1 Hysteretic and direction-dependent atlas gluing

Assumption 4.2 restricts to memoryless compatibility. Real physical systems exhibit hysteresis and direction-dependent overlap structure: first-order phase transitions with

metastable coexistence regions traversed asymmetrically; ferromagnetic memory; rate-independent constitutive memory in materials [Visintin, 1994], [Mielke et al., 2002]. The mathematically clean response is not to modify the overlap structure of the atlas but to change the indexing category: either enlarge the state by internal variables so that the augmented system is memoryless, or work over a category of histories and glue local data along composable trajectories. Neither direction has an established representation-learning framework. Neural modelling of hysteretic systems exists [Chen et al., a], applied in a forward-modelling sense rather than with an atlas-theoretic structural guarantee. A solution would produce an atlas floor theorem analogous to Theorem 4.8 for a hysteresis-compatible structural category — groupoid-valued, fibred-category-valued, or internal-state-augmented.

7.2.2 Overlap-atlas floors at topological contract

Theorem 4.8 is scoped to partition atlases (Def. 3.2), where regions meet only in the negligible set N and the region-assignment function ι is single-valued. For overlap atlases (Def. 3.3), the region-assignment data is a multifunction encoded by the nerve $\mathcal{N}(\{R_\alpha\})$, and the effect of gluing on the dimension floor is not captured by the partition-atlas theorem even at the topological contract.

The phenomenon can raise or lower the floor. Redundant overlaps (nerves that are homotopy-trivial) can leave the floor unchanged or even lower it by supplying extra coordinates usable across regions. Genuinely higher-order overlaps can raise it: the tripod (three intervals glued at a point) has three arms each with $\text{emb} = 1$, overlap dimension $d_B = 0$, and naive floor $\max_\alpha \text{emb}(Q_\alpha) + d_B = 1$, but the glued space requires $D_{\text{ambient}} \geq 2$ since removing the branch point disconnects it into three components while removing a point from any subset of $\mathbb{R}$ leaves at most two.

The overlap-atlas floor theorem at the topological contract would relate D_{ambient} to $\max_\alpha \text{emb}(Q_\alpha)$, d_B , and an overlap-nerve invariant — plausibly the maximum clique number of the nerve, or a higher-homotopy invariant of the nerve combined with attaching-map data. No such theorem is established. This is separate from the non-base-split-encoder problem (§7.3.1) and from the singular-transition excess of Remark 4.9(ii): overlap-atlas effects are global gluing phenomena, not local failures of the base-split hypothesis or of handle regularity.

7.2.3 Stratified-meet compatibility levels and lower-dimensional fibre structure

Definition 3.4 admitted stratified-meet atlases as a third chamber-meeting case alongside partition and open overlap: chambers of varying dimensions meeting along their frontiers, with each chamber carrying its own fibre type. The figure-eight merge — two S^1 chambers entering a junction point at which the fibre is $S^1 \vee S^1$ — is the canonical example. Theorem 4.8 is proved for partition atlases only; its base-split definition does not specify how the latent coordinate splitting interacts with frontier data on lower-dimensional chambers. A floor theorem for stratified-meet atlases is open at the level of both definitions and proofs.

Concretely, two pieces are missing. First, an analogous compatibility-strength axis (§3.3) for stratified meets. The open-overlap axis runs from disjoint (level (a)) through set-theoretic compatibility, isomorphism on overlap, and cocycle (level (d)). For stratified meets there is no open overlap; the analogous data are *frontier-compatibility* prescriptions relating the fibre Q_α on a lower-dimensional chamber to the fibres Q_β on adjacent

higher-dimensional chambers. The figure-eight case is one specific frontier compatibility (wedge sum); others include pushouts, pullbacks, mapping cylinders, and cobordism-type cylinders for Morse-handle attachments. A graded hierarchy of frontier-compatibility levels — with the strongest level corresponding to a coherent stratified-stack reading and the weakest to mere set-theoretic agreement on closures — is open. The constructible-sheaf theory of stratified spaces [Nocera and Volpe, 2023, Ayala et al., 2017] provides the natural machinery; a representation-learning-side translation is not given here.

Second, a base-split contract for stratified meets. The partition-case Definition 4.4 requires the latent linear projection $\pi_{L_B} \circ E$ to recover the base coordinate $\iota_B \circ \pi_B$ on $\mathcal{X} \setminus N$. For a stratified base B with chambers of varying dimensions, the appropriate analogue — whether the same linear projection still works, or whether a stratified family of projections is required, and how the projection interacts with the frontier-compatibility data above — is not given here. Whether the partition-case bound $D_{\text{ambient}} \geq d_B + \sup_b \text{emb}(Q_b)$ extends, with or without modification, to the stratified-meet case is open. The figure-eight case is consistent with such a bound at the naive read — $d_B = 2$ for a tripod, $\max_\alpha \text{emb}(Q_\alpha) = 2$, suggesting $D_{\text{ambient}} \geq 4$ — but neither the right definition of base-split nor the corresponding theorem is supplied. The *boundary structure within N* of the original formulation — separatrices, stable and unstable manifolds of saddles, homoclinic and heteroclinic connections in dissipative systems — is similarly best read as “stratified content the atlas declines to model” rather than as inaccessible structure. Whether such content can be promoted into the atlas as additional lower-dimensional chambers, with their own fibre types and frontier-compatibility prescriptions, is part of the same open modelling question.

7.3 Representation-contract and saturation problems

These problems concern the gap between what the base-split contract enforces and what general atlas-faithful encoding can achieve: non-base-split encoders, saturation of the proved bound, and within-evidence-class identifiability.

7.3.1 Non-base-split encoders and the representation contract

Theorem 4.8 is proved under the base-split hypothesis (Def. 4.4): the latent ambient decomposes as a coordinate product $\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$ in which L_B recovers the base coordinate independently of fibre content. This subsection clarifies what is at stake when that hypothesis fails, and why the hypothesis itself is part of the framework’s design rather than a technical convenience.

The theorem is a floor for a specific representation class. The base-split atlas dimension floor

$$\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q}) \geq d_B + \max_{\alpha} \text{emb}(Q_{\alpha})$$

is the unavoidable latent cost of making the atlas explicit as readable coordinate structure: base coordinates in one factor, fibre content in another, both separately legible. It is not, and is not claimed to be, a floor on every faithful encoding of the underlying total space. A non-base-split encoder may store base and fibre information in the same latent coordinates, using an entangled map that recovers neither coordinate block in isolation but jointly permits region recognition and per-fibre faithfulness. The Möbius band illustrates the phenomenon: it is locally an interval bundle over S^1 but admits no global product splitting; it embeds in $\mathbb{R}^3$, where a base-split representation of the same

underlying space would require $\mathbb{R}^4$. For atlas structures generally, the gap

$$\text{emb}_{\text{top}}(\mathcal{Q}) \leq \text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q})$$

allowed by Def. 4.7 is the price of choosing the base-split contract over an entangled one.

Why the base-split contract is the target class. Open Problem 7.3.1 does not ask whether arbitrary entangled embeddings can sometimes realise an atlas in lower ambient dimension. They can, and such embeddings may still preserve the topology of the total sampled space. The question is whether they preserve the quotient-atlas structure in a stable, interpretable, and diagnostically accessible way. The base-split hypothesis encodes the representational preference that the learned latent space should behave like a coordinate transformation separating base and fibre content, rather than hiding their interaction in an entangled embedding. A non-base-split encoder may be topology-faithful to the total space in the sense of Paper III’s contract hierarchy while failing to be *atlas-faithful* in the stronger structural sense intended here: it can collapse, twist, or mix base and fibre information in a way that preserves global topology but obscures regime recognition, per-chamber quotient comparison, the ability to intervene on base or fibre variables independently, and the diagnostic checks of §5 and §6.

The theorem gives the unavoidable latent cost of making the atlas explicit. If one wants base and fibre to be separately readable, the additive floor is the price. Lower-dimensional entangled encoders are not counterexamples to the theorem; they belong to a different representation contract, one in which atlas structure is no longer a target of the encoding.

Open problems within the base-split contract. Three concrete open problems remain inside the class the theorem covers.

Saturation. Whether the base-split bound is tight — i.e. whether $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q}) = d_B + \max_{\alpha} \text{emb}(Q_{\alpha})$ in general, by an explicit upper-bound construction — is open (§7.3.2).

Singular transitions. A closed-form characterisation of the excess dimension required for partition atlases with singular transitions — coinciding critical values, Morse–Bott degenerate critical submanifolds, high-index attachments outside the $(D_{\text{ambient}} - d_B)$ -fibre budget, stratified-space singularities — is open. Stratified Morse theory [Goresky and MacPherson, 1988] and embedded Morse theory in codimension at least two [Skopenkov, 2008] are the natural conceptual machinery.

Architectural sufficient conditions. Necessary-and-sufficient conditions on neural architectures that force the base-split hypothesis under standard training are open. The architecture-side question links to the broader interpretability programme: which architectures and training procedures naturally produce coordinate-separated representations, and which produce entangled ones? Empirically, gating mechanisms, mixture-of-experts routers, and some attention patterns appear to favour base-split structure; dense fully-connected encoders favour entanglement. A theoretical account is absent.

Open problem outside the base-split contract. Whether there exist nontrivial *intrinsic* atlas floors — lower bounds on $\text{emb}_{\text{top}}(\mathcal{Q})$ over all atlas-faithful encoders, base-split or not, possibly under a contract weaker than Def. 4.3 — is a separate research direction. Within Paper IV’s scope, non-base-split encodings are a different representation class, and finding lower-dimensional entangled encodings of atlas-structured data

is not a way of beating the theorem. The theorem covers the class of representations the framework is designed to produce; the gap to the intrinsic floor is the price of that design choice, not a defect of the proof.

7.3.2 Saturation construction for the base-split atlas floor

Theorem 4.8 is a lower bound. The corresponding upper-bound construction — an explicit base-split atlas-faithful encoder $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ achieving $D_{\text{ambient}} = d_B + \max_{\alpha} \text{emb}(Q_{\alpha})$ for partition atlases with regular Morse-type transitions — is not given in this paper. For specific examples the construction is plausible: each chamber’s local quotient embeds into $\mathbb{R}^{\max_{\alpha} \text{emb}(Q_{\alpha})}$, and handle attachments at regular Morse critical points can be assembled within the $(D_{\text{ambient}} - d_B)$ -dimensional fibre factor over a continuous base embedding $\iota_B : B \hookrightarrow \mathbb{R}^{d_B}$. The double-pendulum atlas with $D_{\text{ambient}} = 5$ (Ex. 4.11) is consistent with the bound, but a rigorous construction, with verification that the assembled encoder is base-split, is open.

The general saturation question — precisely when $\text{emb}_{\text{top}}^{\text{bs}}(\mathcal{Q}) = d_B + \max_{\alpha} \text{emb}(Q_{\alpha})$ versus when excess dimension is required for compatibility-coherent assembly — is plausibly answerable using stratified-embedding tools [Nocera and Volpe, 2023, Natsume, 1980] combined with handlebody-decomposition machinery [Ayala et al., 2017], but neither the formulation nor the proof has been carried out.

7.3.3 Within-equivalence-class resolution

§5.4 established that evidence criteria (E1)–(E3) return evidence-equivalence classes rather than unique atlas hypotheses. Resolving within-class ambiguity requires additional input. Three candidate sources, each with its own caveats: (i) domain knowledge priors (physical-interpretability of specific regime indicators, simplicity preferences for coarser partitions); (ii) auxiliary data not used for selection (transfer tasks, held-out contexts, intervention experiments); (iii) basis-choice or representative-choice criteria from symmetry-reduction and numerical-stability literature [Imakura et al.]. These are partial analogues, not solvers. A rigorous within-class resolution procedure — one that produces a principled unique atlas from an equivalence class given explicit auxiliary input — is open.

7.4 Emergent discovery and learning dynamics

These problems concern the algorithmic and dynamical side of atlas inference: adaptive geometric complexes, training-dynamics implicit bias, generator-class unification, and the three-component structural gap (faithful-summary object, neural base poset/site, training-dynamics compatibility) for emergent discovery.

7.4.1 Atlas-adaptive geometric complexes

Standard persistent homology uses Vietoris–Rips or Čech complexes with uniform scale across the domain. For atlas-structured problems, per-region scale selection — coarser where local topology is simple and sampling is dense, finer where local topology is complex or sampling is sparse — is a natural adaptation analogous to FEM mesh refinement. Cover learning [Scoccola et al., 2025] provides the nearest direct precedent, producing smaller topologically-faithful complexes from data. Adaptive approximation of persistent homology [Sheehy, 2013, Boissonnat and Pritam, 2020] gives computational-efficiency precedent. The missing extensions: local a-posteriori error indicators tied

to atlas-hypothesis-specific refinement criteria; refinement rules that interact well with overlaps and intersections; interleaving-style stability guarantees under non-uniform refinement across regions. The diagnostic promise is that where the atlas hypothesis has misplaced a boundary, the adaptive complex produces visible artifacts at the true boundary — a co-discovery mechanism rather than just a computational efficiency gain. Rigorous development is open.

7.4.2 Atlas-level learning dynamics

A genuinely open question: what happens during training for atlas-structured problems? Does SGD naturally favour certain atlas hypotheses over others? Implicit-bias work on training dynamics — early alignment in two-layer networks [Boursier et al.], region-count simplification [Humayun et al., 2023], gauge-type implicit bias — establishes that training dynamics impose structured implicit preferences, but these results stop short of predicting which atlas hypothesis SGD will favour among several locally admissible ones. This problem borders on Paper 8’s scope (*Neural Realizability of Latent Structure*, forthcoming), which treats the bottom-up theory of which latent structures are naturally realisable by neural parameterisations under training dynamics. A rigorous bridge between atlas selection (§5) and training-dynamics implicit bias would combine Paper IV’s §5 machinery with Paper 8’s stable-migration and accessible-realizability program and remains open.

7.4.3 A unified generator framework covering Lie-type and non-Lie-type regimes

Remark 5.3 noted that Paper VI’s generator specification assumes Lie-group symmetries with free proper action, and that non-Lie regimes (discrete regression structures, chaotic dynamics without continuous symmetry) require analogous but distinct machinery. Paper IV uses Paper VI where it applies and supplements with qualitative specifications elsewhere, without unifying the two. A unified generator framework covering Lie-type, discrete-symbolic, regression-model, and dynamical-symmetry regimes within a single structural category is open. The right formal home is plausibly a category-theoretic generalisation of generator specifications, with $\mathbf{g}$ as an object in a category of structure-specifications, but this is speculative and Paper IV does not pursue it.

7.4.4 Training-compatible faithful-summary objects for neural representations

Component (A) of §6.2 — a faithful neural summary object that is representable as a site, subcanonical (so gluing recovers the represented object), and training-dynamics compatible — is the core mathematical open problem for rigorous emergent discovery. Arya et al. [2025] provides the shape-space analogue. Candidate building blocks exist (local Jacobian spectrum, activation-region adjacency data, centroid-type summaries, persistent features across training perturbations, learned covers). No current candidate has been shown to satisfy all three properties jointly. A demonstrated candidate would convert the emergent-discovery tier from “portfolio of partial diagnostics” to “rigorous procedure with diagnostic guarantees.”

7.4.5 Neural-side base posets and sites

Component (B) of §6.2. The ReLU case admits a natural candidate: the face lattice of the polyhedral activation arrangement [Montúfar et al., 2014, Balestrierio and Baraniuk,

2018], refined by depth. This is the direction Paper 8 (*Neural Realizability of Latent Structure*) develops from first principles. For smooth-activation networks, attention-based models, or hybrid architectures, the natural base object is less clear. Paper IV forward-references Paper 8 for the ReLU case; for other architecture classes, the problem is open.

7.4.6 A realisability-selection bridge between (A), (B), and (C)

Even if each of components (A), (B), (C) of §6.2 is resolved independently, the rigorous emergent-discovery framework requires demonstrating they compose: the faithful summary (A) on the neural base (B) behaves coherently under training dynamics (C). The shape-space analogue composes because shapes are static; neural representations are not. Training-dynamics-compatible sheaf-theoretic descent has no direct analogue in the current mathematical literature, as far as we are aware. This is the highest-level open problem, and plausibly the one whose resolution would close the emergent-discovery tier.

7.5 Beyond quotient atlases

These problems flag directions that exceed the framework’s commitments at the level of the geometric universe itself — bundle-type, higher-symmetric, and path-space redundancies that the integer-dimensional, point-quotient setting cannot capture — and ask what dimension-floor analogues might exist outside that setting.

7.5.1 Dynamics forcing new geometric objects: a direction beyond scope

The eighteen open problems above sit within the geometric universe this framework commits to: integer-dimensional strata, ANR bases, memoryless compatibility, and cocycle-level gluing as the strongest compatibility rung. This universe is adequate for the dynamical systems whose representations we have studied in this series. It is not universal.

A deeper gap is that certain dynamics, when properly formulated, *require* mathematical objects that do not exist in this universe. The prototypical example is quantum field theory. A scalar Klein–Gordon field on Minkowski space can be written classically as a function $\phi : \mathbb{R}^{3,1} \rightarrow \mathbb{R}$, but its quantization — enforcing canonical commutation relations alongside relativistic covariance — forces the introduction of Fock space and creation/annihilation operators, objects that do not exist within the original function space. Fermionic fields go further: spinors do not transform as tensors and are not sections of any tensor bundle; they live on the spin bundle, a double cover of the frame bundle whose existence requires a spin structure on spacetime. The dynamics force the geometric universe to enlarge.

This phenomenon has a direct analogue for quotient representations. The current framework treats redundancy as equivalence of points and collapses equivalence classes to representatives. Some physical and biological systems exhibit redundancy that is not of this form. Natural examples include the following.

Bundle-type redundancy. When the true state space is a principal bundle rather than a quotient — a double cover, a gauge bundle, a spin bundle — the structure cannot be recovered as M/G for a free action. The dimension-floor theorem’s proof relies on a coordinate-projection argument applied to a base-split encoder, which does not transfer to bundle-valued encoders without additional structural data (connection, curvature,

holonomy) governing how the bundle’s total space is to be projected onto a base coordinate.

Higher-symmetric redundancy. Non-Abelian gauge symmetries and certain integrable systems exhibit symmetries forming groupoids or higher groupoids rather than ordinary groups. The atlas compatibility-strength axis of §3.3, topping out at cocycle-level (rung (d)), does not contemplate higher-categorical coherence data. A level-(e) rung allowing weak associativity and higher-cell coherence would be required.

Path-space redundancy in biology. Certain equivalences in biological state spaces — for instance, those induced by developmental trajectories that can pass through the same phenotypic configuration via distinct regulatory routes — are closer to path-space equivalences than point equivalences. These admit homotopy-type rather than set-theoretic quotient descriptions.

In each of these cases, the framework as currently formalised either misses the structure or forces it into a quotient form that discards the information distinguishing the enriched object from its quotient. We flag this as a direction beyond the scope of this series rather than an open problem internal to it. Natural next steps would enlarge the atlas concept from “stratified quotient atlas” to “stratified bundle atlas” or “stratified higher-categorical atlas,” with correspondingly enriched compatibility data. Neither admits a straightforward generalisation of the dimension-floor theorem: the coordinate-projection argument of §4.3 requires a base-split structure that bundle-valued and higher-categorical encoders generally lack, and whether an analogous floor exists for bundle-valued or higher-categorical faithful encoders is itself an open question.

The philosophical point is that the framework’s integer-dimensional, point-quotient universe is not neutral. It is a modelling commitment whose adequacy depends on whether the dynamics under study admit a quotient description at all. For dynamics that do not — gauge fields, fermionic systems, certain biological state spaces — the correct mathematical response is to enlarge the universe, not to force the dynamics into a universe that cannot contain them.

7.5.2 A dimension-floor analogue beyond the quotient universe

The atlas dimension floor (Theorem 4.8) depends essentially on the coordinate-projection argument in its proof (§4.3, step 2), which requires the encoder to be base-split (Def. 4.4). This argument is specific to encoders mapping into a Euclidean target with a coordinate splitting that recovers the base, and does not extend to the generalised settings described in §7.5.1: bundle-valued encoders do not project onto a base coordinate in the required sense without additional structural data, and higher-categorical atlases do not admit a base dimension of the right type for a coordinate splitting to apply.

Problem. Does there exist a dimension-floor analogue for faithful encoders valued in any of the following enriched targets: (i) principal bundles over a stratified base, (ii) groupoid-equivariant spaces, (iii) path-space-type homotopy quotients? If so, what form do the base-dimension term and any additional coherence terms take, and how does the base-split hypothesis (Def. 4.4) generalise to these targets?

Candidate routes include a Chern-class-type obstruction in the principal-bundle case; an Euler-characteristic-weighted count of groupoid object classes for the groupoid case; a total-persistence analogue over the path space for the homotopy case. None of these has been worked out, and it is unclear whether any admits a floor with the same structural form as Theorem 4.8 (per-region floor plus base-dimension floor under a base-split

condition) or whether the two-term decomposition itself breaks outside the quotient setting.

A proof that no such floor exists — for instance, a construction of arbitrarily-low-dimensional faithful bundle-valued encoders for a fixed base — would be equally valuable, as it would establish that the integer-dimensional, point-quotient universe is not merely a modelling commitment but an essential one for the floor to hold. This problem is categorically different from the seventeen preceding open problems, which all sit within the framework’s current commitments; it asks whether the central theorem of this paper has any analogue at all once those commitments are relaxed.

A Mathematical Background for ML Readers

This appendix collects mathematical notions from outside the standard ML curriculum that the body of Paper IV assumes. Each entry is meant to be enough for a reader to bootstrap on the body without consulting external references; readers familiar with geometric topology can skip it. The appendix is not a textbook substitute — proofs live elsewhere (see Appendix B for the load-bearing classical embedding theorems).

A.1 Topological foundations

Hausdorff space. A topological space in which any two distinct points have disjoint open neighbourhoods. This is the standard “separation” property: it ensures limits of sequences are unique and that singletons $\{x\}$ are closed. All metric spaces are Hausdorff. The line with two origins (mentioned in §3.5 as a non-separated example) fails Hausdorff at the doubled point. Throughout Paper IV the input space $\mathcal{X}$ is required to be Hausdorff; the latent ambient $\mathbb{R}^{D_{\text{ambient}}}$ is automatically Hausdorff. *Reference:* Munkres, *Topology* (2000), Ch. 17.

Compact space. A space in which every open cover has a finite subcover. For metric spaces, equivalently: every sequence has a convergent subsequence. The Heine–Borel characterisation says a subset of $\mathbb{R}^n$ is compact iff it is closed and bounded. Compactness is the topological version of “fits in a finite region,” and it makes continuous maps well-behaved: a continuous map from a compact space has compact (hence closed and bounded) image, attains its supremum if real-valued, and is uniformly continuous if the codomain is a metric space. Paper IV requires $\mathcal{X}$ to be compact; for the worked examples this means the realised data support, not the full physical phase space.

Paracompact space. Every open cover admits a locally finite refinement (each point has a neighbourhood meeting only finitely many sets in the refinement). Strictly weaker than compactness: $\mathbb{R}^n$ is paracompact but not compact. Required by classical embedding theorems (e.g. Whitney) to ensure partitions of unity exist. All metric spaces are paracompact; this is enough for Paper IV’s purposes since the spaces in question are metrisable. *Reference:* Munkres Ch. 41.

Continuous map. In metric spaces: small input changes produce small output changes (ε – δ). In topological spaces: the preimage of any open set is open. The encoder E in Paper IV is required to be continuous; this is what makes “the latent shape” $E(\mathcal{X})$ a meaningful geometric object rather than a scattered cloud.

Embedding. A continuous injective map $f : X \rightarrow Y$ that is a homeomorphism onto its image — i.e., $f^{-1} : f(X) \rightarrow X$ exists and is also continuous. Two equivalent formulations: (a) f is continuous, injective, and open onto $f(X)$; (b) f is a continuous bijection $X \rightarrow f(X)$ such that X and $f(X)$ are homeomorphic. Topological embedding is the weakest sense; bi-Lipschitz, smooth, and isometric embeddings (Paper III §3.4) are stronger refinements that preserve more structure. The word “embedding” alone in Paper IV means topological embedding unless qualified.

Connected component. A maximal connected subspace. Two points lie in the same component iff some connected subspace contains both. $\mathbb{R}^n$ has one component (it is connected); a finite discrete set with k elements has k components; the disjoint union of S^1 and a point has two. The §7.3.1 discussion of base-split vs. entangled encoders turns on whether the latent shape’s component structure encodes regime identity.

Quotient map. A surjective map $\pi : X \rightarrow Y$ such that Y has the finest topology making π continuous: $U \subseteq Y$ is open iff $\pi^{-1}(U) \subseteq X$ is open. Equivalently, Y is obtained from X by an equivalence relation, with π sending each x to its equivalence class. The quotient maps $\pi_b : \pi_B^{-1}(b) \rightarrow Q_b$ in Paper IV Def. 3.1 are quotient maps in this sense — they collapse each fibre to its quotient object. *Reference:* Munkres Ch. 22.

A.2 Dimension theory

Dimension is a more delicate concept in topology than in linear algebra. Several inequivalent notions of dimension exist for general topological and metric spaces, and Paper IV uses different ones in different places. The key distinctions matter because the floor theorem’s d_B term is *embedding* dimension, not covering dimension, and the two genuinely differ.

Topological covering dimension ($\dim_{\text{cov}}$). The minimum integer n such that every open cover of the space has a refinement in which no point lies in more than $n + 1$ sets. For “reasonable” spaces this matches intuition: $\dim_{\text{cov}}(\mathbb{R}^n) = n$, $\dim_{\text{cov}}(S^n) = n$, $\dim_{\text{cov}}(\text{finite discrete set}) = 0$. Defined intrinsically — doesn’t depend on an ambient space. *Reference:* Engelking, *Theory of Dimensions Finite and Infinite* (1995).

Topological embedding dimension (emb_{top}). The minimum integer d such that the space embeds (as a topological subspace, see §A.1) into $\mathbb{R}^d$. Always at least the covering dimension, but can be strictly larger. Standard examples: $\text{emb}_{\text{top}}(S^1) = 2$ (the circle doesn’t fit in a line) but $\dim_{\text{cov}}(S^1) = 1$; $\text{emb}_{\text{top}}(K_5) = 4$ for the complete graph on five vertices but $\dim_{\text{cov}}(K_5) = 1$. The Menger–Nöbeling theorem (Appendix B.1) gives the bound $\text{emb}_{\text{top}}(X) \leq 2 \dim_{\text{cov}}(X) + 1$ for separable metric spaces. The theorem of Paper IV bounds latent dimension in terms of $d_B = \text{emb}_{\text{top}}(B)$ and $\text{emb}(Q_\alpha)$, both embedding dimensions.

Hausdorff dimension ($\dim_H$). A metric notion taking arbitrary nonnegative real values, defined via r -cover scaling: $\dim_H(X) = \inf\{s : \mathcal{H}^s(X) = 0\}$ where $\mathcal{H}^s$ is the s -dimensional Hausdorff measure. Captures fractal geometry; agrees with covering dimension on tame spaces. The Cantor set has $\dim_H = \log 2 / \log 3 \approx 0.63$ but $\dim_{\text{cov}} = 0$. Paper IV restricts to integer-dimension cases (Assumption 4.1); non-integer-dimension atlases are a named open problem.

Doubling dimension. The minimum δ such that every metric ball $B(x, r)$ can be covered by at most 2^δ balls of radius $r/2$. Captures local volume-growth rate. Coincides with vector-space dimension for $\mathbb{R}^n$ (where $\delta = n$); sharper than both Hausdorff and covering dimensions for irregular spaces. Used in Paper III §3.4 for bi-Lipschitz embedding floors and Paper IV Remark 3.5.

Assouad dimension. A uniform variant of doubling: the smallest s such that all balls at all scales admit a uniform δ -doubling bound. Always at least the Hausdorff dimension. Paper IV Remark 3.5 uses Assouad dimension as the natural invariant in the non-Archimedean (ultrametric) case: a compact ultrametric with infinite Assouad dimension cannot bi-Lipschitz embed into any finite-dimensional Euclidean space.

A.3 Manifolds and stratified spaces

Paper IV’s body treats the input space $\mathcal{X}$ as compact Hausdorff but not necessarily a manifold. The fibre objects Q_b are typically manifolds; the assembled space $\mathcal{X}$ may be stratified.

Topological n -manifold. A second-countable Hausdorff space that is locally homeomorphic to $\mathbb{R}^n$: every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$. The number n is the manifold’s dimension.

Closed manifold. Compact, without boundary. Examples: $S^n, T^n, \mathbb{CP}^n$. The double-pendulum fibre objects are closed 3-manifolds.

Smooth manifold. A topological manifold equipped with a smooth atlas — overlapping coordinate charts whose transition maps are smooth. Allows differentiation. Whitney’s theorem (Appendix B.2) says every smooth n -manifold embeds smoothly in $\mathbb{R}^{2n}$.

Orientable manifold. Admits a consistent choice of “positive direction” (formally: an atlas with all transition maps having positive Jacobian determinant; or, a non-vanishing top-degree differential form). Examples: spheres, tori, $\mathbb{CP}^n$. Counter-examples: Möbius band, Klein bottle, $\mathbb{RP}^2$. Closed orientable 3-manifolds embed in $\mathbb{R}^5$ (Hirsch); they cannot embed in $\mathbb{R}^3$.

Stratified space. A space X partitioned into disjoint subsets $\{X_i\}$ called *strata*, each a manifold of possibly different dimension, such that the closure of each stratum is a union of strata (the “frontier condition”). Smooth and topological stratifications carry additional regularity (Whitney conditions, Thom regularity, etc.). The double-pendulum phase space is stratified by energy regime: the open regimes are 4-manifold strata, the critical-energy hypersurfaces are lower-dimensional strata. Paper IV’s atlas formalism treats stratified inputs via the negligible set N and the chamber decomposition. *Reference:* Goresky–MacPherson, *Stratified Morse Theory* (1988); for a modern atlas-theoretic treatment, Ayala–Francis–Tanaka (2017).

Connected sum $M \# N$. For closed orientable n -manifolds M and N : remove a small open ball from each, glue the resulting boundaries by an orientation-reversing homeomorphism. The result is a closed orientable n -manifold. The notation $\#_2(S^2 \times S^1) =$

$(S^2 \times S^1) \# (S^2 \times S^1)$ used in Paper IV §2.2 refers to two copies of $S^2 \times S^1$ joined at a point. *Reference:* Hatcher, *Algebraic Topology* (2002), Ch. 1.

Handle attachment. The operation that builds an n -manifold M' from an n -manifold-with-boundary M by gluing $D^k \times D^{n-k}$ along $S^{k-1} \times D^{n-k}$, where D^k is the closed k -disk and S^{k-1} its boundary. The integer k is the handle's *index*. A 0-handle adds a ball; a 1-handle adds a 1-dimensional bridge; a 2-handle fills in a 2-disk along a circle in the boundary; an n -handle caps off. Morse theory says the topology of a sublevel set $f^{-1}((-\infty, E])$ changes by handle attachment of index equal to the Morse index of the critical point as E crosses a critical value. *Reference:* Milnor, *Lectures on the h-Cobordism Theorem* (1965).

Handlebody. A manifold built up by attaching handles to a ball. A genus- g 3-dimensional handlebody is D^3 with g 1-handles attached. Its boundary is the closed orientable surface of genus g . Used in Appendix B.4 for the embedding $\#_2(S^2 \times S^1) \hookrightarrow \mathbb{R}^4$ as the boundary of a 4-dimensional genus-2 handlebody.

A.4 Morse theory

The double-pendulum example uses Morse theory to pass from a smooth function (the Hamiltonian) to a topological stratification of phase space (the fibre topology sequence). ML readers without geometric topology background may not have encountered Morse theory.

Morse function. A smooth function $f : M \rightarrow \mathbb{R}$ on a manifold whose critical points (where $df = 0$) are all *nondegenerate* — the Hessian matrix at each critical point is invertible. Nondegenerate critical points are isolated. Morse functions form an open dense subset of all smooth functions in any reasonable topology — i.e., a generic smooth function is Morse. *Reference:* Milnor (1965); Audin–Damian, *Morse Theory and Floer Homology* (2014).

Morse index. At a nondegenerate critical point, the number of negative eigenvalues of the Hessian. Equals the dimension of the maximal subspace on which f has a local maximum at the critical point. For an n -dimensional manifold, the index is between 0 (a local minimum) and n (a local maximum); index-1 critical points are saddles.

Critical value. A value $c \in \mathbb{R}$ such that $f^{-1}(c)$ contains a critical point. The set of critical values of a Morse function on a compact manifold is finite. Between consecutive critical values, all level sets $f^{-1}(c)$ are mutually diffeomorphic; crossing a critical value modifies the topology by handle attachment.

Sublevel set vs. regular level set. $f^{-1}((-\infty, E])$ is the *sublevel set*; $f^{-1}(E)$ for E a regular value (not critical) is a *regular level set* and is itself a smooth manifold of dimension $\dim M - 1$. Morse theory is naturally formulated for sublevel sets — they grow by handle attachment as E crosses critical values. The double-pendulum example in §2.2 reports the topology of the regular level sets $\Sigma_E = H^{-1}(E)$, not the sublevel sets $H^{-1}((-\infty, E])$. The two are related: regular level sets are the boundaries of consecutive sublevel sets between critical values, and the regular-level-set topology changes by *surgery* (cut out $S^{k-1} \times D^{n-k-1}$, glue in $D^k \times S^{n-k-2}$) corresponding to the handle attachment on the sublevel side.

Morse–Bott function. A generalisation allowing critical *submanifolds* rather than isolated critical points. The Hessian is required to be non-degenerate transverse to the critical submanifold. A perfectly symmetric system (e.g. a free group action with Morse-symmetric Hamiltonian) typically has Morse–Bott rather than Morse critical structure. Mentioned in Remark 4.9 as a singular-transition class outside Theorem 4.8’s regular-handle-attachment scope.

Stratified Morse theory. The extension of Morse theory to stratified spaces, accommodating critical points that lie on lower-dimensional strata. Goresky–MacPherson’s monograph (1988) is the standard reference; the theory is invoked in §7.3.1 as the natural conceptual machinery for the singular-transition open problem.

A.5 Algebraic-topology tools

Persistent homology. A multi-scale invariant of a filtered topological space. Given a sequence of nested spaces $X_0 \subseteq X_1 \subseteq \dots$, persistent homology tracks how homology classes (loops, voids) appear and disappear across the filtration; the output is a *persistence diagram* or *barcode* recording birth and death scales. In TDA applied to point clouds: build a Vietoris–Rips or Čech filtration parameterised by scale ε , extract the persistent Betti numbers. The diagnostic portfolio of §6 invokes PH to estimate fibre topology from data without committing to a specific filtration choice. *References:* Edelsbrunner–Harer, *Computational Topology* (2010); Carlsson, “Topology and data” (2009).

Nerve of a cover. Given a cover $\{U_\alpha\}$ of a space X , the *nerve* $\mathcal{N}(\{U_\alpha\})$ is the simplicial complex with one k -simplex for each nonempty $(k+1)$ -fold intersection $U_{\alpha_0} \cap \dots \cap U_{\alpha_k}$. Captures the combinatorial structure of how cover sets overlap. Under regularity hypotheses (the nerve theorem: contractible intersections), the nerve has the same homotopy type as the covered space. Used in Paper IV §3.5 and §7.2.2.

Cocycle condition. For a cover with overlap data $\phi_{\alpha\beta} : U_\beta \rightarrow U_\alpha$ on each pairwise overlap, the cocycle condition is $\phi_{\alpha\gamma} = \phi_{\alpha\beta} \circ \phi_{\beta\gamma}$ on each triple overlap $U_\alpha \cap U_\beta \cap U_\gamma$, plus $\phi_{\alpha\alpha} = \text{id}$. Cocycle data on a cover is the input to descent: it specifies how to glue local objects into a global one. The compatibility-axis level (d) in §3.3 is the cocycle condition.

Homotopy equivalence. Two spaces X, Y are homotopy-equivalent if continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ exist with $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$ (homotopic to identity). Homotopy-equivalent spaces share homology, homotopy groups, and most topological invariants but may have different dimensions. A line and a point are homotopy-equivalent (via deformation retraction), as are an annulus and a circle.

Absolute neighbourhood retract (ANR). A space X such that whenever X is embedded as a closed subset of a normal space Y , there is a neighbourhood U of X in Y that retracts continuously onto X . Most “tame” spaces are ANRs: manifolds, finite CW complexes, finite-dimensional locally compact metric ANRs. Mentioned in Paper IV for context; not required by the current proof of Theorem 4.8 (the ANR hypothesis was dropped in revision).

A.6 Special metric structures

Bi-Lipschitz map. A map $f : (X, d_X) \rightarrow (Y, d_Y)$ between metric spaces with constants $c_1, c_2 > 0$ such that $c_1 d_X(a, b) \leq d_Y(f(a), f(b)) \leq c_2 d_X(a, b)$ for all a, b . The *distortion* is the smallest ratio c_2/c_1 ; a bi-Lipschitz map with distortion 1 is an isometry. Bi-Lipschitz maps preserve connectedness, completeness, Hausdorff dimension, and doubling structure, but not Euclidean structure.

Isometric map. Distance-preserving: $d_Y(f(a), f(b)) = d_X(a, b)$. The strongest metric contract. Isometric embeddings of Riemannian manifolds into Euclidean space are governed by Nash–Kuiper (C^1 , generic) and Nash–Günther (C^∞ , with high target dimension). The Nash–Kuiper C^1 -isometric embedding of a flat torus into $\mathbb{R}^3$ exists despite the Hartman–Nirenberg obstruction to C^2 isometric embedding.

Ultrametric. A metric satisfying the *strong triangle inequality* $d(x, z) \leq \max(d(x, y), d(y, z))$, which is strictly stronger than the standard triangle inequality. Ultrametric spaces have a hierarchical structure: the open balls form a tree, and any two open balls are either disjoint or one contains the other. The p -adic numbers $\mathbb{Q}_p$ carry a canonical ultrametric. Bi-Lipschitz embedding of an ultrametric into Euclidean space is impossible unless the ultrametric is uniformly disconnected. Used in Paper IV §7.1.5 for the non-Archimedean contract family.

Snowflake metric. For (X, d) a metric space and $\alpha \in (0, 1]$, the *snowflake* (X, d^α) — same set, distance raised to power α . The triangle inequality still holds because $\alpha \leq 1$. Snowflaking can transform a space without good Euclidean embedding properties into one that does embed: every doubling space admits a bi-Lipschitz embedding of its α -snowflake into Euclidean space for some $\alpha < 1$ (Assouad’s theorem; Naor and Neiman 2012 sharpened the dimension dependence). Used in Paper III §3.4 and Paper IV Remark 3.5.

A.7 Specific named spaces

p -adic numbers $\mathbb{Q}_p$. The completion of the rationals $\mathbb{Q}$ with respect to the p -adic absolute value: $|p^k m/n|_p = p^{-k}$ when $\gcd(m, p) = \gcd(n, p) = 1$. Numbers divisible by high powers of p are p -adically *small*. $\mathbb{Q}_p$ is a field, complete, locally compact, totally disconnected, and ultrametric. Used in Paper IV Remark 3.5 as a target class for the non-Archimedean contract.

Complex projective space $\mathbb{CP}^n$. The space of complex 1-dimensional subspaces of $\mathbb{C}^{n+1}$, equivalently $S^{2n+1}/U(1)$. A compact complex manifold of complex dimension n (real dimension $2n$). Carries the canonical Fubini–Study metric and symplectic form. Used in Paper IV §7.1.3 for symplectic-target atlas floors (Tischler / Gromov).

Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. A specific principal S^1 -bundle realising S^2 as the base and S^3 as the total space. The simplest non-trivial principal bundle, and the model for Paper II’s Hopf-fibration single-symmetry experiments. Mentioned briefly in Paper IV §1.

Torus T^n . The n -fold product $S^1 \times S^1 \times \cdots \times S^1$. A closed orientable n -manifold. T^2 embeds in $\mathbb{R}^3$ (donut); T^3 embeds in $\mathbb{R}^4$ but not $\mathbb{R}^3$. Used in Paper IV §2.2 as the

high-energy fibre of the double pendulum and as the configuration space of the double pendulum itself.

B Proofs of Classical Results

This appendix records the precise statements of the classical embedding theorems invoked in the body, with proof sketches for the load-bearing cases. The point is verification rather than novelty: the body cites these results in passing; readers should not have to chase external references to confirm what the citations claim. Proof sketches are provided where the proof is short enough to be informative; for the deeper theorems we give a structural outline and a primary reference. The final subsection (§B.9) addresses a different question: are any of Paper IV’s results genuine, or are they recast classical theorems?

B.1 Menger–Nöbeling embedding theorem

Theorem (Menger 1928, Nöbeling 1931). *Every separable metric space X of topological covering dimension $\dim_{\text{cov}}(X) \leq n$ admits a topological embedding into $\mathbb{R}^{2n+1}$.*

Proof sketch. Consider the space $C(X, \mathbb{R}^{2n+1})$ of continuous maps $X \rightarrow \mathbb{R}^{2n+1}$ in the supremum metric, with X compact for concreteness. The maps that fail to be embeddings are those that fail injectivity on some pair of points, or fail homeomorphism onto image at some point. Both failure conditions define meagre subsets of $C(X, \mathbb{R}^{2n+1})$ provided the target dimension is high enough. The general-position calculation is: for X of covering dimension n , the set $\{(x, y) \in X \times X : x \neq y\}$ has covering dimension at most $2n$, and a generic continuous map into $\mathbb{R}^{2n+1}$ separates pairs of distinct points. The Baire category theorem then gives a residual set of embeddings, hence at least one. Compactness of X promotes the resulting injective continuous map to a homeomorphism onto its image. $\square$

Where used in Paper IV. Remark 3.5 (Archimedean contract family); Cor. 4.10 (single-quotient reduction, where $|A| = 1$ and the result reduces to the per-fibre Menger–Nöbeling bound); §7.1.2 stronger-contract programme.

Reference. Menger (1928), *Dimensionstheorie*. Nöbeling (1931), “Über eine n -dimensionale Universalmenge im $\mathbb{R}^{2n+1}$.” Modern treatment: Engelking (1995), *Theory of Dimensions Finite and Infinite*, Theorem 1.10.4.

B.2 Whitney embedding theorem

Theorem (Whitney 1944, strong form). *Every smooth n -manifold M admits a smooth embedding into $\mathbb{R}^{2n}$.*

Theorem (Whitney 1936, weak form). *Every smooth n -manifold admits a smooth embedding into $\mathbb{R}^{2n+1}$.*

Proof sketch (weak form). Let M be compact (the non-compact case is by exhaustion). Cover M by finitely many coordinate charts $\{(U_i, \varphi_i)\}_{i=1}^k$ with $\varphi_i : U_i \rightarrow \mathbb{R}^n$. Let $\{\rho_i\}$ be a partition of unity subordinate to the cover. Define

$$f : M \rightarrow \mathbb{R}^{nk} \times \mathbb{R}^k, \quad x \mapsto (\rho_1(x)\varphi_1(x), \dots, \rho_k(x)\varphi_k(x), \rho_1(x), \dots, \rho_k(x)).$$

This map is smooth, injective, and an immersion. The image lies in a high-dimensional Euclidean space; a generic linear projection onto $\mathbb{R}^{2n+1}$ remains an embedding by Sard’s theorem. $\square$

The strong form (target $\mathbb{R}^{2n}$) requires the Whitney trick to remove self-intersections of an immersion — a subtler argument involving the geometry of double points. We cite rather than reproduce.

Where used in Paper IV. Remark 3.5 (smooth contract); §7.1.2 stronger contracts; §B.4 (DP fibre embeddings).

Reference. Whitney (1944), “The self-intersections of a smooth n -manifold in $2n$ -space.” Modern treatment: Hirsch (1976), *Differential Topology*, Ch. 1–2.

B.3 Haefliger refinements

Theorem (Haefliger 1961, 1962). *Let M be a closed smooth n -manifold with $n \geq 2$. If M is k -connected (i.e. $\pi_i(M) = 0$ for $0 < i \leq k$), then M embeds smoothly in $\mathbb{R}^{2n-k}$, provided $2n - k \geq n + 3$.*

Haefliger’s theorem strengthens Whitney’s $\mathbb{R}^{2n}$ bound by trading connectedness of the manifold for reduced target dimension. The condition $2n - k \geq n + 3$, equivalent to $k \leq n - 3$, is required because the proof uses the Whitney trick, which needs at least 3 codimensions to operate.

Proof structure. Haefliger’s argument constructs a smooth immersion using obstruction theory, then removes self-intersections using a parametrised version of the Whitney trick. The connectedness hypothesis $\pi_i = 0$ for $i \leq k$ ensures that the obstruction classes vanish at each stage of an inductive construction.

Where used in Paper IV. Remark 3.5 (smooth contract); §7.1.2 stronger-contract programme.

Reference. Haefliger (1962), “Plongements différentiables de variétés dans variétés.” For an account in modern language: Skopenkov (2008), *Embedding and knotting of manifolds in Euclidean spaces*, Surveys in Contemporary Mathematics.

B.4 Embedding dimensions of the four DP fibre types

Ex. 4.11 asserts $\text{emb}(Q_\alpha) = 4$ for each of S^3 , $S^2 \times S^1$, $\#_2(S^2 \times S^1)$, T^3 . This subsection gives the upper bound (each embeds in $\mathbb{R}^4$) and lower bound (none embeds in $\mathbb{R}^3$) per case.

Lower bound: no closed orientable 3-manifold embeds in $\mathbb{R}^3$.

Lemma. *If M is a connected closed 3-manifold (compact, no boundary, non-empty) and $f : M \rightarrow \mathbb{R}^3$ is a topological embedding, then $f(M)$ would be both open and closed and bounded in $\mathbb{R}^3$, which is impossible.*

Proof. The image $f(M)$ is compact, hence closed and bounded in $\mathbb{R}^3$. The image is also open: by invariance of domain (Brouwer 1912), a continuous injective map from an n -manifold into an n -manifold is open. So $f(M)$ is open and closed in $\mathbb{R}^3$. Connectedness of $\mathbb{R}^3$ forces $f(M) \in \{\emptyset, \mathbb{R}^3\}$; non-emptiness rules out $\emptyset$, and unboundedness of $\mathbb{R}^3$ rules out $f(M) = \mathbb{R}^3$ (since $f(M)$ is bounded). Contradiction. $\square$

This rules out $\mathbb{R}^3$ as a target for any of the four fibre types: each is connected, closed, and orientable.

Upper bound for S^3 .

$S^3 = \{x \in \mathbb{R}^4 : \|x\| = 1\}$ is the standard unit 3-sphere; the inclusion is smooth and an embedding. So $\text{emb}(S^3) \leq 4$, and combined with the lower bound, $\text{emb}(S^3) = 4$.

Upper bound for $S^2 \times S^1$.

$S^2 \times S^1$ bounds a 4-manifold-with-boundary: it is the boundary of $S^2 \times D^2$ (where D^2 is the closed disk). Embed $S^2 \times D^2 \hookrightarrow \mathbb{R}^4$ by realising S^2 as the unit sphere in $\mathbb{R}^3 = \mathbb{R}^3 \times \{0\} \subset \mathbb{R}^4$ and taking a tubular neighbourhood (a small disk bundle) around it; for an unknotted $S^2 \subset \mathbb{R}^4$ the normal bundle is trivial, so the tubular neighbourhood is $S^2 \times D^2$. Then $S^2 \times S^1 = \partial(S^2 \times D^2) \hookrightarrow \mathbb{R}^4$ as the boundary of the embedded 4-manifold-with-boundary. $\square$

Upper bound for $\#_2(S^2 \times S^1)$.

The connected sum $\#_2(S^2 \times S^1) = (S^2 \times S^1) \# (S^2 \times S^1)$ is the boundary of a *genus-2 four-dimensional handlebody* — a 4-ball with two 1-handles attached. Concretely: let H_2 denote the result of attaching two thickened arcs (each diffeomorphic to $D^3 \times [0, 1]$) to a 4-ball D^4 along disjoint pairs of disks in the boundary $S^3 = \partial D^4$. This H_2 is a 4-manifold-with-boundary; its boundary ∂H_2 is $\#_2(S^2 \times S^1)$, since each 1-handle attachment to D^4 produces an $S^2 \times S^1$ boundary contribution (cf. Milnor 1965). The genus-2 four-dimensional handlebody H_2 admits a standard smooth embedding in $\mathbb{R}^4$: take D^4 as the unit 4-ball, and realise the two 1-handles as small thickened arcs in $\mathbb{R}^4$ joining disjoint pairs of disks on $\partial D^4 = S^3$, chosen with sufficient codimension that the thickenings can be made disjoint and embedded. Hence $\#_2(S^2 \times S^1) = \partial H_2 \hookrightarrow \mathbb{R}^4$. $\square$

Upper bound for T^3 .

Embed T^2 into $\mathbb{R}^4$ as the standard product torus

$$(\theta, \phi) \mapsto (\cos \theta, \sin \theta, \cos \phi, \sin \phi).$$

This is a smooth codimension-two embedding. The normal bundle is trivial: at each (θ, ϕ) the two normal directions are spanned by the unit radial vectors $(\cos \theta, \sin \theta, 0, 0)$ in the first $\mathbb{R}^2$ -factor and $(0, 0, \cos \phi, \sin \phi)$ in the second, which form a global trivialisation since the splitting $\mathbb{R}^4 = \mathbb{R}^2 \oplus \mathbb{R}^2$ is fixed independently of (θ, ϕ) . A sufficiently small tubular neighbourhood is therefore diffeomorphic to $T^2 \times D^2$, and its boundary is $T^2 \times S^1 = T^3$, giving the embedding $T^3 \hookrightarrow \mathbb{R}^4$. $\square$

Conclusion. $\text{emb}(Q_\alpha) = 4$ for each of S^3 , $S^2 \times S^1$, $\#_2(S^2 \times S^1)$, T^3 .

B.5 Hirsch’s general 3-manifold theorem

Theorem (Hirsch 1961). *Every closed orientable 3-manifold admits a smooth embedding into $\mathbb{R}^5$.*

This theorem is stronger than what §B.4 establishes (which gave $\mathbb{R}^4$ for the four specific DP fibres) in that it holds for *all* closed orientable 3-manifolds; weaker in target dimension.

Proof structure. The proof has two ingredients. First, every closed orientable 3-manifold is parallelisable (Stiefel 1936): the tangent bundle is trivial. Second, parallelisability allows the immersion-theoretic obstruction to embedding in $\mathbb{R}^5$ to vanish; the codimension-2 obstruction theory (Haefliger–Hirsch) then upgrades the immersion to an embedding when codimension is at least 2 and the manifold is parallelisable. Hirsch’s contribution is the embedding theorem; Stiefel’s parallelisability result is the input.

Where used in Paper IV. Ex. 4.11 cites this as the general fallback bound; the specific DP fibre types do better (embed in $\mathbb{R}^4$).

Reference. Hirsch (1961), “The imbedding of bounding manifolds in Euclidean space.” Modern treatment: Hirsch (1976), *Differential Topology*, §5–6.

B.6 Hurewicz dimension-lowering inequality

Theorem (Hurewicz 1927). *Let $f : X \rightarrow Y$ be a continuous map between separable metric spaces. Then $\dim_{\text{cov}}(X) \leq \dim f + \dim_{\text{cov}}(Y)$, where $\dim f := \sup_{y \in Y} \dim_{\text{cov}}(f^{-1}(y))$ is the supremum of fibre dimensions.*

This is the foundational result that “dimension cannot drop below fibre dimension plus base dimension” for continuous maps. Paper IV Theorem 4.8 is closely related: it bounds the latent dimension below by base dimension plus per-fibre embedding dimension (under base-split). The key distinction is that Hurewicz uses *covering* dimension on both sides while Theorem 4.8 uses *embedding* dimension. We address this distinction in §B.9.

Proof sketch. The proof uses that covering dimension is characterised by extension properties: $\dim_{\text{cov}}(Z) \leq n$ iff every continuous map $A \rightarrow S^n$ from a closed subset $A \subseteq Z$ extends to all of Z . Given continuous $f : X \rightarrow Y$ with fibres of dimension $\leq k$ and $\dim_{\text{cov}}(Y) \leq m$, one constructs extensions through f by combining (i) extensions on fibres and (ii) extensions across the base, using the fibre structure of f . The full argument is technical and relies on the *countable sum theorem* for covering dimension: a countable union of closed subsets each of dimension $\leq k$ has dimension $\leq k$. $\square$

Where used in Paper IV. §4 lead-in cites Hurewicz as one of the three classical dimension-lowering heuristics that Theorem 4.8 sharpens into an embedding-dimension contract.

Reference. Hurewicz–Wallman (1948), *Dimension Theory*, Princeton University Press, Theorem VI.7. Modern treatment: Engelking (1995), *Theory of Dimensions Finite and Infinite*, Ch. 4.

B.7 Natsume’s stratified realisation theorem

Theorem (Natsume 1980). *Every paracompact abstract stratified set of (top) dimension n admits a topological embedding into $\mathbb{R}^{2n+1}$.*

This is the stratified analogue of Menger–Nöbeling. It gives an *upper bound* on the embedding dimension of stratified spaces; Paper IV Theorem 4.8 gives a *lower bound* on the latent dimension required for atlas-faithful encoding. The two are in different directions and are not equivalent.

Proof structure. Natsume’s argument adapts the general-position embedding strategy to the stratified setting. One embeds the strata individually into Euclidean targets of appropriate dimension, then glues using regularity hypotheses on the stratification (in Natsume’s case, frontier conditions and locally cone-like structure). The gluing is the technical heart; embedding the strata is standard.

Structural remark. A reader might ask: doesn’t Natsume’s theorem give Theorem 4.8 directly? It does not. Natsume’s bound is on the *minimum dimension at which embedding becomes possible*. Theorem 4.8 bounds the *latent ambient dimension* — which is the dimension of the latent space, fixed by the architecture’s exposed-layer width — for an encoder that satisfies an additional structural requirement (base-split). The two quantities can in principle differ by an arbitrary amount: the minimal embedding dimension of an atlas-structured input space is at most $2n + 1$ by Natsume, but the minimal *base-split* atlas-faithful encoding dimension can be larger.

Where used in Paper IV. §7.1.2 stronger-contract programme; Remark 3.5 for context on stratified targets.

Reference. Natsume (1980), “The realization of abstract stratified sets,” *Kodai Mathematical Journal* 3(1), 1–7.

B.8 Snowflake / Assouad embedding

Theorem (Assouad 1983; sharpened by Naor–Neiman 2012). *Let (X, d) be a metric space with doubling dimension $\delta < \infty$, and let $\alpha \in (0, 1)$. The snowflaked metric space (X, d^α) admits a bi-Lipschitz embedding into $\mathbb{R}^N$ where $N \leq C(\delta, \alpha)$ depends only on δ and α .*

The Naor–Neiman refinement gives explicit dimension dependence: for α bounded away from 1, $N \leq O(\delta)$; the dimension explodes as $\alpha \rightarrow 1$.

Proof structure (Assouad’s original). Use a hierarchical decomposition of X at geometric scales $r_k = r_0 \cdot 2^{-k}$. At each scale, cover X by balls of radius r_k ; the doubling property bounds the cover size. Construct an embedding using the cover structure plus a Lipschitz partition of unity, with the snowflake exponent α controlling how distortion propagates across scales. The bound on N is multiplicative across scales but independent of the metric beyond doubling.

Where used in Paper IV. Remark 3.5 (bi-Lipschitz contract family); §7.1.2 bi-Lipschitz atlas floors.

References. Assouad (1983), “Plongements lipschitziens dans $\mathbb{R}^n$.” [Naor and Neiman \[2012\]](#).

B.9 Literature-recast check: is Theorem 4.8 just Hurewicz?

The reviewer-style question this subsection addresses: *is Paper IV’s Theorem 4.8 a genuine new result, or is it a recast of a known classical theorem in different language?*

The honest answer has two parts.

Part 1: At the level of dimensional bookkeeping, Theorem 4.8 is closely related to Hurewicz dimension-lowering. Both bound the dimension of a total space from below by base dimension plus fibre dimension. Both treat the base map (Hurewicz: arbitrary continuous f ; Theorem 4.8: the base map π_B) as the projection that defines fibres. If one replaced “embedding dimension” with “covering dimension” throughout the statement of Theorem 4.8, the resulting bound $\dim_{\text{cov}}(\mathcal{X}) \leq \dim_{\text{cov}}(B) + \sup_b \dim_{\text{cov}}(\pi_B^{-1}(b))$ is exactly Hurewicz. So at the level of *covering* dimension, the inequality is classical.

Part 2: Theorem 4.8 is not Hurewicz, for two reasons. First, Theorem 4.8 is about *embedding* dimension, not covering dimension. The inequality $\text{emb}_{\text{top}}(B) + \sup_b \text{emb}(Q_b) \leq D_{\text{ambient}}$ does not follow from Hurewicz: Hurewicz gives $\dim_{\text{cov}}(\mathcal{X}) \leq \dim_{\text{cov}}(B) + k$ where k is fibre covering dimension, but the relationship between covering and embedding dimensions is one-sided ($\text{emb}_{\text{top}}(X) \leq 2 \dim_{\text{cov}}(X) + 1$ from Menger–Nöbeling, with no reverse bound). The theorem’s content includes a coordinate-projection construction in the latent ambient that establishes the embedding-dimension inequality directly, not via covering dimension.

Second, Theorem 4.8 is a statement about *encoders*, not about the abstract space $\mathcal{X}$. The bound says: if $E : \mathcal{X} \rightarrow \mathbb{R}^{D_{\text{ambient}}}$ is a base-split atlas-faithful encoder, then $D_{\text{ambient}} \geq d_B + \sup_b \text{emb}(Q_b)$. The hypothesis (base-split) is a structural requirement on the encoder, not a geometric property of $\mathcal{X}$. Hurewicz gives no such conditional encoder bound; it is a statement about the relationship between dimensions of source, target, and fibres of an arbitrary continuous map.

Substantive contribution of Theorem 4.8. Given parts 1 and 2: Theorem 4.8 is novel in its *combination* — embedding-dimension bookkeeping + base-split contract + atlas-structured input — but the *proof technique* (coordinate projection plus per-fibre embedding) is classical. This is why the proof is short. Paper IV’s substantive contribution is therefore not the bound itself, which is essentially forced by the combination of definitions, but the *framework* in which the bound becomes meaningful: the atlas formalism, the base-split representation contract, the three-tier separation, and the eighteen open problems. The theorem is the framework’s load-bearing minimal example, not its peak technical achievement.

This honest assessment is consistent with the position taken in §1’s “What this paper is and is not” paragraph: the substantive contribution is the framework; whether the base-split bound is also a general atlas-faithful bound — whether non-base-split encoders can ever achieve strictly lower latent dimensions — is open (§7.3.1).

Other classical results Paper IV does not recast. A few other candidate classical results worth checking:

- Menger–Nöbeling (§B.1) gives an upper bound on embedding dimension from covering dimension. Theorem 4.8 gives a lower bound on a coordinate-split encoder dimension. Different direction.
- Whitney / Haefliger (§B.2, §B.3) give upper bounds on smooth embedding dimensions for manifolds. Theorem 4.8’s setting is topological, not smooth, and gives a lower bound. Different direction.
- Hirsch (§B.5) is about specific embeddings of closed orientable 3-manifolds into $\mathbb{R}^5$. Used in Ex. 4.11 as a reference for the per-case embedding dimensions, not as the substantive content.
- Natsume (§B.7) gives an upper bound on stratified-set embedding dimension (the structural-remark distinction in §B.7 applies).
- Snowflake / Assouad (§B.8) is about bi-Lipschitz embeddings of doubling metric spaces. Different contract; flagged as relevant for §7.1.2 but not the basis of Theorem 4.8.

The literature-recast verdict: *Theorem 4.8 is essentially classical at the level of the inequality, but the framing — base-split atlas-faithful encoders for atlas-structured representation problems — is new, and the theorem is the natural minimal floor in that framework.* The framework is the contribution; the theorem is what makes the framework concrete.

C Novelty and Contribution Attribution

This appendix provides (i) a row-by-row attribution of the intellectual contributions in Paper IV and (ii) a primary-contribution statement, critical-prior-work subsection, existence-vs-constructive check, and AI-tools disclosure. The methodology and the prompt that produced this appendix appear in Appendix D.

C.1 Context

Paper IV was developed across approximately twelve working sessions in April–May 2026. The primary drafting tool was Claude Sonnet 4.6 and Claude Opus 4.7 (Anthropic). Editorial review and reviewer-style critique were occasionally cross-checked using ChatGPT 5.4 (OpenAI) and Gemini 3 Pro (Google), but no AI other than Claude contributed substantively to the body of the paper.

The recognition that atlas structure was needed at all came from Paper VII [Farkas, 2026g]: the double pendulum’s energy stratification and the Color Fantasy ship-hotel HVAC regime structure exposed the limits of the single-quotient theory of Papers I–V. The quotient atlas as a generalisation, the three-tier separation between given-partition, learned-partition, and emergent-discovery problems, the compatibility-strength axis, and the eighteen open problems organising the field were the author’s framings.

The base-split-as-representation-contract resolution in §7.3.1 was driven by the author with the operative formulation proposed almost verbatim (*“Lower-dimensional entangled encoders are not counterexamples; they belong to a different representation contract”*). The author rejected the earlier framing that treated non-base-split encoders as evidence the theorem was weak. The Color Fantasy continuous-base correction in Remark 4.13 was the author’s recognition that earlier-draft framing as a discrete-base atlas conflated qualitative regime labels with the underlying continuous physical variables.

The three-failure-modes restructure in Remark 4.9 — splitting the earlier compatibility-coherence term $c(\mathcal{Q})$ into three named classes (non-base-split, singular transitions, overlap atlases) — was a joint development in which the author recognised the bundling problem and the AI assisted in articulating the three-way separation. The decomposition is now load-bearing for §7.3.1, §7.3.2, and §7.2.2.

C.2 Attribution Table

The attribution table below uses the weight scheme from Appendix D: **0** (not applicable), **+** (substantive contribution), **++** (primary responsibility; this party could have produced this result independently of the others). **AI** **++** is reserved exclusively for results volunteered without being asked; executing a given direction, writing prose, or supplying literature when prompted are **+**, not **++**.

Result	Character	Lit.	Author	AI
<i>Framework and formalisations</i>				
Quotient atlas as generalisation of single-quotient framework: the atlas concept itself, motivated by Paper VII’s empirical work	novel frame-work	+	++	0
Three-tier separation: given-partition, learned-partition, and emergent-discovery as distinct theoretical regimes	novel frame-work	0	++	0
Quotient atlas formal definition (Def. 3.1): base, base map, regime labels, chambers, fibre objects, contract–target pairs	novel formalisation	0	++	+
Partition vs. overlap atlas distinction (Defs. 3.2, 3.3): single-valued ι versus multifunction Λ encoded by the nerve	novel formalisation	0	++	+
Compatibility-strength axis with four levels (a)–(d) (§3.3), mapped onto the descent-strength hierarchy of sheaf and stack theory	novel frame-work + classical mapping	+	++	+
Branched contract hierarchy (Rmk 3.5): Archimedean (linear chain), non-Archimedean ultrametric, structure-faithful, with explicit pairwise-incomparability claims and the canonical (C, T) pairs	novel reframing	+	++	+
Categorical reading (§3.5, optional): partition as coproduct, overlap as colimit over the nerve, compatibility levels as sheaf conditions, base-split vs. scheme-theoretic separatedness distinction	novel framing	+	++	+

Result	Character	Lit.	Author	AI
Base-split atlas-faithful encoder (Def. 4.4): global linear coordinate factorisation $\mathbb{R}^{D_{\text{ambient}}} = L_B \oplus L_F$ recovering the base independently of fibre content; explicit exclusion of merely-functional base-recoverability	novel formalisation + contract framing	0	++	+
<i>The dimension floor and consequences</i>				
Theorem 4.8: base-split atlas dimension floor $D_{\text{ambient}} \geq d_B + \sup_b \text{emb}(Q_b)$, with chamber-form corollary	classical book-keeping + novel encoder framing	++	++	+
Three failure modes restructure (Rmk 4.9): the earlier additive term $c(\mathcal{Q})$ split into non-base-split, singular transitions, and overlap atlases as genuinely distinct phenomena with different mathematical origins	novel reframing	0	++	+
Ex. 4.11: double pendulum atlas $D_{\text{ambient}} \geq 5$ floor, with per-case verification of $\text{emb}(Q_\alpha) = 4$ for S^3 , $S^2 \times S^1$, $\#_2(S^2 \times S^1)$, and T^3	classical (4-manifold theory) + novel atlas application	++	+	+
Ex. 4.12 and Rmk 4.13: Color Fantasy continuous-base reading and earlier-draft correction; $d_B = 2$ from joint cabin-bay temperature space; year-phase scope	novel reframing	0	++	+
§7.3.1: non-base-split encoders as a different representation contract, not a defect of the bound; explicit framing of base-split as encoding a representational preference for coordinate-separated structure	novel reframing	0	++	+
<i>Learned-partition and emergent-discovery tiers</i>				
Admissible atlas family (Def. 5.1) and the joint-selection problem; coupling of $\{g_\alpha\}$ and the regime-indicator class $\mathcal{R}$ as joint rather than independent choices, with evidence-equivalence-class consequence	novel formalisation, extends Paper VI	+	++	+
Three evidence criteria (E1)–(E3) and evidence-equivalence classes (§§5.3–5.4): per-regime faithfulness, region-stratified topological invariants, transition-signature diagnostics	novel frame-work	+	++	+

Result	Character	Lit.	Author	AI
Three-component structural gap (A)–(C) for emergent discovery (§6.2): faithful-summary object, neural base poset/site, training-dynamics compatibility, drawn by analogy with the Arya–Curry–Mukherjee sheaf-theoretic shape-space construction	novel synthesis	+	++	+
Diagnostic portfolio of five tool classes (§6.3): topological/sheaf, dynamical/spectral, info-theoretic/causal/mechanistic-interpretability, multimanifold/symmetry-discovery, perturbation-robustness; presented as a parallel menu rather than a unified pipeline, with readiness assessments	novel framing	++	++	+
Contract-level considerations at the emergent tier (§6.5): the empirical work of Papers II, V, and VI is already operating at the structure-faithful contract (physics losses) rather than the topological one; observation framed as reframing of existing literature, not derivation	novel reframing	+	++	+
<i>Open problems</i>				
Eighteen open problems organised from most concrete to most speculative (§7); twelve foundational, three contract-family, three structural-gap; with prior-work positioning where relevant	novel synthesis	0	++	+
§§7.5.1–7.5.2: dynamics forcing geometric objects beyond the quotient universe (bundle-type, higher-symmetric, path-space redundancy) and the dimension-floor-analogue question for enriched targets; explicitly flagged as categorically different from the seventeen preceding open problems	research direction	+	++	+
<i>AI-assisted execution (all under author direction)</i>				
Prose drafting from author outlines and notes; LaTeX typesetting; citation-format hygiene	execution	0	0	+

Result	Character	Lit.	Author	AI
Literature search and identification of related work (e.g. AtlasD, Arya–Curry–Mukherjee, Bhat et al., Nocera–Volpe, AyalaFrancisTanaka) when directed	execution	0	0	+
Per-case verification and writeup of Ex. 4.11 embedding bounds (Appendix B.4); proof execution for Theorem 4.8 from author’s two-step outline	execution	+	0	+

No row in this paper meets the criterion of an AI ++. Every contribution attributed to the AI was in response to a direction from the author or constituted execution (prose, typesetting, literature search, formalisation of an author-supplied structure). The closest candidates considered for AI ++ were the three-failure-modes restructure (Rmk 4.9) and the Arya–Curry–Mukherjee parallel for the structural gap (§6.2); both were ultimately classified as joint contributions in which the author identified the gap and the AI assisted in articulating or surfacing the relevant parallel, rather than unsolicited contributions volunteered without being asked.

C.3 Primary Contribution

The substantive contribution of Paper IV is the framework: the quotient atlas as a structural object generalising the single-quotient theory of Papers I–V; the distinction between base, regime labels, and fibre objects; the compatibility-strength axis ranking gluing data by descent strength; the three-tier separation between given-partition, learned-partition, and emergent-discovery problems; the diagnostic portfolio organising five tool classes for partial progress on the emergent tier; and the eighteen open problems that catalogue what remains. Theorem 4.8 (the base-split atlas dimension floor) is the framework’s load-bearing minimal example, not its peak technical achievement: its inequality is essentially classical at the level of bookkeeping (closely related to Hurewicz dimension-lowering; see §B.9), and the proof is short. What makes the theorem novel is the encoder-conditional, embedding-dimension framing under the base-split representation contract — none of which appears in the classical literature.

The paper does not solve emergent atlas discovery. It identifies the three-component structural gap (faithful-summary object, base poset/site, training-dynamics compatibility) by analogy with the sheaf-theoretic shape-space construction of Arya et al. [2025], and presents a portfolio of partial diagnostic tool classes rather than a unified pipeline. Whether the diagnostic portfolio composes into a rigorous emergent-discovery procedure remains open (§§7.4.4–7.4.6).

C.4 Critical Prior Work

Paper IV builds most directly on four works from the present series:

- Farkas [2026a] (Paper I): the non-certification result for standard training criteria. Paper IV extends this to the per-region failure mode that arises in atlas-structured problems: task performance can be achieved by encoders that pass within-region

evaluation while assigning different unfaithful encodings in different regions, a failure mode undetectable from within-region evaluation alone.

- [Farkas \[2026c\]](#) (Paper III): the single-quotient embedding-dimension floor. Paper IV’s Theorem 4.8 is the atlas-level generalisation of Proposition 3.1 of Paper III, recovered as Cor. 4.10 when $|A| = 1$ and the base is a single point.
- [Farkas \[2026f\]](#) (Paper VI): generator specifications and the admissible quotient class $\mathcal{Q}_{\text{cl}}(\mathfrak{g}, \mathcal{X})$. Paper IV’s admissible atlas family (Def. 5.1) is the atlas-level extension, with regularity hypotheses required per chamber rather than globally and supplemented by non-Lie per-regime specifications where Paper VI’s hypotheses do not hold.
- [Farkas \[2026g\]](#) (Paper VII): empirical validation of the framework on the phase cylinder, Hopf fibration, double pendulum, and Color Fantasy. The recognition that atlas structure was needed came from Paper VII’s empirical work, with the double pendulum’s energy stratification and the Color Fantasy regime structure providing the two motivating cases. The DP topology sequence and the CF regime structure are inherited from Paper VII under the stated normalisation, not re-derived here.

External works the framework draws on most directly include [Vakil \[2024\]](#) for the categorical reading and the base-split versus scheme-theoretic-separatedness distinction; [Goresky and MacPherson \[1988\]](#) for stratified Morse theory underlying the singular-transition open problem; [Arya et al. \[2025\]](#) for the sheaf-theoretic faithful-summary construction motivating the three-component structural gap; and [Bhat et al. \[2025\]](#) (AtlasD) as close-parallel work treating per-region structural discovery in an atlas framework (Rmk 5.6).

C.5 Existence vs. Constructive Check

Theorem 4.8 is a lower bound on the latent dimension required for a base-split atlas-faithful encoder. It is not a constructive result: it does not exhibit any encoder achieving the bound, and saturation is named as Open Problem 7.3.2. The Ex. 4.11 statement that the double pendulum atlas has $D_{\text{ambient}} \geq 5$ is consistent with Paper VII’s empirically observed floor of $D_{\text{ambient}} = 5$ but does not by itself constitute a proof of saturation: an explicit base-split atlas-faithful encoder realising $D_{\text{ambient}} = 5$ is not constructed here.

The §6 diagnostic portfolio is similarly non-constructive at the framework level: it organises tool classes by readiness rather than supplying an algorithm. §6.4 explicitly states that the portfolio enables a workflow but does not yet produce a rigorous certification of emergent atlas discovery; closing that loop is an open problem (§§7.4.4–7.4.6).

The paper does not conflate existence and learnability. Where existence-only results are stated, they are flagged as such, and the corresponding learnability or constructive question is named in §7. The within-evidence-equivalence-class ambiguity of §5.4 is similarly an identifiability statement, not a learnability statement; resolving it requires additional input (§7.3.3).

C.6 AI Tools

Tool	Version	Phase	Role
Claude Sonnet	4.6 (Anthropic, Apr. 2026)	Drafting, revision	Prose drafting, formalisation execution, citation-formatting, reviewer-style critique under author direction
Claude Opus	4.7 (Anthropic, May 2026)	Late revision, appendices	Appendix A and Appendix B drafting under author outlines; literature-recast check (§B.9); per-case embedding writeups; Appendix C and D under author direction
ChatGPT	5.4 (OpenAI, Apr. 2026)	Cross-check	Occasional reviewer-style critique used to surface phrasing or framing weaknesses; no body content originated here
Gemini 3 Pro	Google, Apr. 2026	Cross-check	Occasional reviewer-style critique; no body content originated here

Each working session was conducted in a fresh conversation context. The seed prompt, working conventions, and locked decisions from prior sessions were re-established at each session start; nothing was held in implicit AI memory across sessions. The author reviewed every draft change, accepted or revised proposed text directly, and intervened to correct mathematical or framing errors. Generative AI outputs are non-deterministic and cannot be exactly reproduced; the paper should be evaluated on its mathematical and intellectual content alone. The intellectual genesis of all novel contributions belongs to the first author; the AI’s role was expressive and executional, not generative. The first author takes full scientific and intellectual responsibility for all claims in the paper, including those rows where the AI is marked +.

D The Attribution Prompt and Worked Example

D.1 The Prompt

The following prompt was used to generate Appendix C. It can be adapted for any AI-assisted academic paper.

Read the full paper and the full conversation history before making any attributions. The paper alone cannot establish who originated an idea or whether the AI acted on a direction or volunteered something unprompted. The conversation history is the primary evidence for Author vs. AI weight assignments. Attributions made without it are unreliable.

Identify every distinct intellectual unit: named theorems, definitions, conjectures, remarks, open problems, syntheses of existing results, reframings of existing phenomena, and novel connections. Each is a separate row. Do not include purely pedagogical illustrations.

For each row, write a short descriptive label in the Character column — one

to four words, honest and descriptive. A row that is genuinely two things can say so.

Assign weights across three columns — **Lit.** (prior literature), **Author**, **AI** — using:

- **0** not applicable
- **+** substantive contribution
- **++** primary responsibility; this party could have produced this result independently

Calibration: Author **++** when the idea originated with the human author and would not have appeared without their specific input. AI **++** reserved exclusively for results volunteered without being asked — not for executing a given direction, not for writing prose, not for identifying a theorem when prompted. Supplying a theorem when asked is **+**. Proposing its relevance unprompted is **++**.

Produce the table, then write: a **Primary Contribution** paragraph stating the paper’s main contribution plainly; a **Critical Prior Work** subsection naming the two or three papers the work builds on most directly; a check for any conflation between existence results and constructive or learnability results; and an **AI Tools** subsection listing each tool, its version, phase, and role, closing with a statement that the human author takes full responsibility for all claims.

D.2 Worked Example: Paper IV

The full output produced by applying the prompt above to Paper IV — attribution table, primary contribution, critical prior work, existence-vs-constructive check, and AI-tools subsection — appears in Appendix C. We do not reproduce it here. A few observations on what the prompt produced for this particular paper:

Empty AI ++ column. As in Paper I and Paper III, the AI **++** column is empty across all rows. The closest candidates were the three-failure-modes restructure (Rmk 4.9) and the parallel to the sheaf-theoretic shape-space construction motivating the structural gap (§6.2). Both were classified, after review of the conversation history, as joint contributions in which the author identified the gap and the AI assisted in articulating or surfacing the relevant parallel. Neither met the standard of *volunteered without being asked*. This is the correct outcome for a paper in which the author drove the framework, the structural decisions, and the open-problem catalogue.

Mixed Lit / Author rows. Several rows carry both Lit **++** and Author **++**. This reflects the paper’s design as a framework applying classical mathematics rather than novel mathematics: Theorem 4.8 relies on classical bookkeeping (Hurewicz, Menger-Nöbeling style projection arguments) but is itself a new statement under a new contract; the diagnostic portfolio surveys mature tool classes (persistent homology, Takens embedding, mechanistic-interpretability methods) but organises them under a framework-specific readiness scheme. The double-mark is honest: the literature and the author both have substantive responsibility.

Execution rows. The bottom three rows (prose drafting, literature search, per-case verification) carry AI + with both Lit and Author at 0 or +. This reflects the genuine division of labour: the AI executed mechanical tasks under direction. These rows are included for completeness and honesty about the working process, not because the underlying tasks would be considered intellectual contributions in the absence of AI assistance.

What the prompt does not capture. The prompt produces a row-level attribution. It does not capture the iterative cross-checking process by which framings were refined — many rows passed through several revisions in dialogue with the AI before the formulation in the body of the paper was reached. The conversation history records this; the table abstracts away from it. Readers who want a finer-grained account should consult the working logs, which the author maintains.

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